

Spacelike reduction of the gravitational topological terms and associated helicity densities

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Abstract

Helicity in gravity is a multifaceted concept, and several inequivalent definitions exist in the literature. In this work, we reduce the topological Pontryagin and Nieh-Yan terms to a constant time spacelike hypersurface. Remarkably, upon applying the SVT decomposition at the linearized level, we obtain four distinct helicity densities. The helicity density arising from the Nieh-Yan term is termed the spin-1 and spin-2 gravitomagnetic helicity density, depending on whether it originates from vector or tensor modes. Meanwhile, the helicity density arising from the Pontryagin term is termed the spin-1 and spin-2 gravito-current helicity density, according to the corresponding mode contributions. We study the mode and multipole expansions of these helicity functionals and apply them to leading order Newtonian two-body systems, weak field boosted Kerr, a locally defined slowly varying gyratonic pp-wave model, and a linearized gravitational Hopfion.

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1 Introduction

Helicity plays related but inequivalent roles in different areas of physics. In the representation theory of massless particles, it labels irreducible representations of the little group [1]. In free Maxwell theory, optical helicity can be identified with the Noether charge associated with electromagnetic duality rotations [2, 3]. For a barotropic ideal fluid, the velocity field $\mathbf{v}$ and vorticity $\boldsymbol{\omega}$ define the fluid helicity $\int d^3x \mathbf{v} \cdot \boldsymbol{\omega}$ [4, 5]. Its magnetohydrodynamic counterpart is the magnetic helicity $\int d^3x \mathbf{a} \cdot \mathbf{b}$ [5–8] where $\mathbf{a}$ is the magnetic potential and $\mathbf{b}$ the magnetic field. On a domain with a boundary, the gauge invariance of magnetic helicity requires suitable boundary conditions or the use of relative helicity [9]. Gravitoelectromagnetism, i.e., the analogy between linearized gravity and electromagnetism in the weak field and slow motion regime, has been reviewed and developed in [10, 11]. Within this framework, a gravitomagnetic helicity construction was proposed in [12]. Related parity odd spin-1 and spin-2 quantities arise in duality based descriptions of linearized gravity [13–16]. Electromagnetic analogues include Lipkin’s zilch and optical chirality [17–19], while

parity odd tensor structures associated with the gravitational Pontryagin density also appear in parity violating gravity and cosmology [20, 21].

The present work is motivated by two recent developments. The first concerns the radiative electromagnetic helicity flux at future null infinity $\mathcal{I}^+$ [22–24]. This quantity should be distinguished from the magnetic, electric, and optical helicities defined as charges on spacelike hypersurfaces. The second development involves the comparison of the asymptotic descendants of the gravitational Pontryagin and Nieh-Yan densities [25]. Under the standard Bondi-Sachs falloff conditions [26–28], that analysis finds that the complete Pontryagin Chern-Simons three-form alone does not reproduce the proposed radiative spin-2 helicity flux [29], whereas the teleparallel Nieh-Yan descendant can yield a parity odd $\epsilon^{AB}C_{AC}\partial_u C_B{}^C$ -type flux. Here C_{AB} denotes the Bondi shear and u is retarded time.

A useful observation is that several of the helicity densities relevant to the present work arise from the descent of exact characteristic classes. In electromagnetism, the Abelian Pontryagin density, or equivalently the second Chern character, satisfies

$$da \wedge da = d(a \wedge da), \quad (1.1)$$

and the pullback of $a \wedge da$ to a spacelike hypersurface gives the magnetic helicity. In gravity,

$$R^a{}_b \wedge R^b{}_a = dQ_3, \quad (1.2)$$

where Q_3 is the gravitational Chern-Simons three-form. Likewise, the Nieh-Yan density is exact [30],

$$N_{\text{NY}} = d(e^a \wedge T_a) = T^a \wedge T_a - e^a \wedge e^b \wedge R_{ab}. \quad (1.3)$$

Here e^a denotes the vielbein (or coframe) one-form, while T^a and $R^a{}_b$ denote the torsion and curvature two-forms associated with the spin connection $\omega^a{}_b$, respectively. In teleparallel equivalent of general relativity (TEGR), $R^a{}_b(\omega) = 0$, since $\omega^a{}_b$ is a flat teleparallel spin connection. This should not be confused with the generally non-vanishing curvature of the Levi-Civita connection. The Pontryagin and Nieh-Yan densities can enter chiral anomaly relations [31–34]. The Nieh-Yan coefficient, however, generally depends on the regularization prescription or ultraviolet scale and should not be regarded as universal in the same sense as the Pontryagin contribution.

In this paper, we investigate the pullback of the Nieh-Yan boundary form to a spacelike hypersurface in the linearized TEGR. Using the scalar-vector-tensor (SVT) decomposition [35], we show that, modulo codimension-two surface terms, the resulting bulk term separates into spin-1 and spin-2 contributions. The spin-1 contribution has the same form as gravitomagnetic helicity, up to convention dependent normalization, whereas the spin-2 contribution is closely related to terms appearing in duality based descriptions of linearized gravitational waves [13, 14]. To our knowledge, the latter has not previously been isolated as the spin-2 sector of the spatial Nieh-Yan descendant.

We also perform the corresponding reduction of the gravitational Pontryagin density. Its spatial descendant again separates, modulo codimension-two terms, into spin-1 and spin-2 bulk sectors. Their derivative-weighted parity odd structure is analogous to current helicity and to Lipkin’s zilch or optical chirality [17–19]. Since these bulk expressions are written entirely in terms of the SVT gauge invariant variables, they are invariant under linearized diffeomorphisms. Their separation from the surface terms, as well as their conservation properties, nevertheless depends on the equations of motion and the imposed falloff conditions. We therefore use the term “helicity” for these parity odd quadratic functionals without assuming that every one of them is an independently conserved Noether charge.

The structure of this paper is organized as follows. Section 2 presents the SVT decomposition of the gravitational field. Using the six gauge invariant modes, Section 3 defines four helicity densities by reducing the Pontryagin and Nieh-Yan terms to a constant time hypersurface. Section 4 elaborates on the mode expansions of these densities and Section 5 focus on their multipole expansions.

These formulations are then applied to two-body systems, weak field boosted Kerr, a slowly varying gyratonic pp-wave model and a gravitational Hopfion before concluding.

2 Setup

In this section, we will follow [35] to review the SVT decomposition of the gravitational field in flat spacetime. The SVT decomposition in a broader context can be found in [36, 37]. We will focus on linearized gravity where the metric $g_{\mu\nu}$ is considered as a small deformation of the Minkowski metric $\eta_{\mu\nu}$

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} \quad (2.1)$$

where $\eta_{\mu\nu} = \text{diag}(-, +, +, +)$ and the magnitude of the gravitational field $h_{\mu\nu}$ is rather small $|h_{\mu\nu}| \ll 1$. The linearized coordinate transformation of $h_{\mu\nu}$ is

$$\delta_\xi h_{\mu\nu} = -\partial_\mu \xi_\nu - \partial_\nu \xi_\mu = -2\partial_{(\mu} \xi_{\nu)} \quad (2.2)$$

where ξ^μ denotes the infinitesimal coordinate transformation $x^\mu \rightarrow x'^\mu = x^\mu + \xi^\mu$ and its indices are lowered by $\eta_{\mu\nu}$. In an arbitrary gauge, we decompose $h_{\mu\nu}$ according to the irreducible representation of the spatial SO(3) group

$$h_{00} = 2\phi, \quad (2.3a)$$

$$h_{0i} = \beta_i + \partial_i \gamma, \quad (2.3b)$$

$$h_{ij} = h_{ij}^{\text{TT}} + \frac{1}{3}H\delta_{ij} + \partial_{(i}\epsilon_{j)} + (\partial_i\partial_j - \frac{1}{3}\delta_{ij}\nabla^2)\lambda. \quad (2.3c)$$

We use normalized symmetrization and Laplacian as follows¹

$$X_{(i}Y_{j)} \equiv \frac{1}{2}(X_iY_j + X_jY_i), \quad \nabla^2 = \delta^{ij}\partial_i\partial_j. \quad (2.4)$$

Here $H = \delta^{ij}h_{ij}$ is the spatial trace of h_{ij} while h_{ij}^{TT} is a symmetric, transverse and traceless spin-2 mode in the sense that

$$h_{ij}^{\text{TT}} = h_{ji}^{\text{TT}}, \quad \partial^i h_{ij}^{\text{TT}} = 0, \quad \delta^{ij}h_{ij}^{\text{TT}} = 0. \quad (2.5)$$

Therefore, only two of the components of h_{ij}^{TT} are independent. Similarly, β_i and ϵ_i are transverse spin-1 modes due to the constraints

$$\partial^i \beta_i = 0, \quad \partial^i \epsilon_i = 0. \quad (2.6)$$

Thus each spin-1 mode contributes two independent components. Combining with the four SO(3) scalar ϕ, γ, H, λ modes, the number of the independent components is exactly ten

$$2 + 2 \times 2 + 4 = 10. \quad (2.7)$$

A parallel decomposition of the vector field ξ^μ is

$$\xi_\mu = (A, B_i + \partial_i C) \quad (2.8)$$

¹A similar anti-symmetrization will be defined as

$$X_{[i}Y_{j]} \equiv \frac{1}{2}(X_iY_j - X_jY_i).$$

where B_i is transverse

$$\partial^i B_i = 0. \quad (2.9)$$

It follows that the following combinations are gauge invariant at the linearized level

$$\Phi = -\phi + \dot{\gamma} - \frac{1}{2}\ddot{\lambda}, \quad \Theta = \frac{1}{3}(H - \nabla^2 \lambda), \quad \Xi_i = \beta_i - \frac{1}{2}\dot{\epsilon}_i, \quad h_{ij}^{\text{TT}}. \quad (2.10)$$

Note that Ξ_i is transverse $\partial^i \Xi_i = 0$. In the following, we will refer h_{ij}^{TT} as the spin-2 modes, Ξ_i the spin-1 modes. The remaining two quantities Φ, Θ are gauge invariant scalar modes. We assume that all perturbations fall off sufficiently rapidly at spatial infinity so that the SVT decomposition is unique and harmonic zero modes can be excluded. Given the linearized metric $h_{\mu\nu}$, the gauge invariant modes are obtained via the formula

$$\Phi = -\frac{1}{2}h_{00} + \nabla^{-2}\partial^i \dot{h}_{0i} - \frac{3}{4}\nabla^{-4}\partial^i \partial^j \ddot{h}_{ij} + \frac{1}{4}\nabla^{-2}\ddot{h}_{ii}, \quad (2.11a)$$

$$\Theta = \frac{1}{2}\left(h_{ii} - \nabla^{-2}\partial^i \partial^j h_{ij}\right), \quad (2.11b)$$

$$\Xi_i = \hat{P}_i^j \left(h_{0j} - \nabla^{-2}\partial^k \dot{h}_{jk}\right), \quad (2.11c)$$

$$h_{ij}^{\text{TT}} = \hat{\Pi}_{ij,kl} h_{kl}. \quad (2.11d)$$

where the projector is defined with the inverse Laplacian

$$\hat{P}_{ij} = \delta_{ij} - \frac{\partial_i \partial_j}{\nabla^2}, \quad \hat{\Pi}_{ij,kl} = \frac{1}{2} \left(\hat{P}_{ik} \hat{P}_{jl} + \hat{P}_{il} \hat{P}_{jk} - \hat{P}_{ij} \hat{P}_{kl} \right). \quad (2.12)$$

Unless stated otherwise, the spatial SVT decomposition is defined on a complete constant time slice $V \simeq \mathbb{R}^3$ of the inertial Minkowski foliation. All perturbations and admissible infinitesimal diffeomorphisms are assumed to satisfy the same regularity and falloff conditions at spatial infinity. The inverse Laplacian is defined by the Green function that vanishes at spatial infinity,

$$(\nabla^{-2} f)(\mathbf{x}) = -\frac{1}{4\pi} \int_{\mathbb{R}^3} d^3 \mathbf{y} \frac{f(\mathbf{y})}{|\mathbf{x} - \mathbf{y}|}, \quad (2.13)$$

or, equivalently, for nonzero Fourier modes,

$$\widehat{\nabla^{-2} f}(\mathbf{k}) = -\frac{\hat{f}(\mathbf{k})}{|\mathbf{k}|^2}, \quad \mathbf{k} \neq 0, \quad \nabla^{-4} = (\nabla^{-2})^2.$$

No independent zero or harmonic modes are retained. They are fixed by regularity and the boundary conditions at spatial infinity. Consequently, the term "gauge invariant" below means invariant under infinitesimal diffeomorphisms that preserve these conditions. Since the SVT projectors contain inverse Laplacians, the resulting modes are nonlocal fields defined on the entire slice $\mathbb{R}^3$. Whenever an idealized example does not satisfy the assumed global falloff conditions, we interpret the displayed SVT modes only as local or asymptotic expressions, supplemented implicitly by an appropriate infrared completion.

The linearized Christoffel symbol is

$$\Gamma_{00}^0 = -\dot{\phi}, \quad (2.14a)$$

$$\Gamma_{0i}^0 = -\partial_i \phi, \quad (2.14b)$$

$$\Gamma_{ij}^0 = \frac{1}{2}\dot{h}_{ij} - \partial_{(i} \beta_{j)} - \partial_i \partial_j \gamma, \quad (2.14c)$$

$$\Gamma_{00}^i = \dot{\beta}_i + \partial_i \dot{\gamma} - \partial^i \phi, \quad (2.14d)$$

$$\Gamma_{0j}^i = \frac{1}{2}\dot{h}_{ij} + \frac{1}{2}(\partial_j\beta_i - \partial_i\beta_j), \quad (2.14e)$$

$$\Gamma_{jk}^i = \frac{1}{2}(\partial_j h_{ki} + \partial_k h_{ji} - \partial_i h_{jk}) \quad (2.14f)$$

while the Riemann tensor is

$$R_{0i0j} = \partial_i\partial_j\Phi + \partial_{(i}\dot{\Xi}_{j)} - \frac{1}{2}\ddot{h}_{ij}^{\text{TT}} - \frac{1}{2}\delta_{ij}\ddot{\Theta}, \quad (2.15a)$$

$$R_{0ijk} = \frac{1}{2}(\partial_k\dot{h}_{ij}^{\text{TT}} - \partial_j\dot{h}_{ik}^{\text{TT}}) + \frac{1}{2}\partial_i(\partial_j\Xi_k - \partial_k\Xi_j) + \frac{1}{2}(\delta_{ij}\partial_k\dot{\Theta} - \delta_{ik}\partial_j\dot{\Theta}), \quad (2.15b)$$

$$R_{ijkl} = \frac{1}{2}(\partial_k\partial_j h_{il}^{\text{TT}} + \partial_l\partial_i h_{jk}^{\text{TT}} - \partial_k\partial_i h_{jl}^{\text{TT}} - \partial_l\partial_j h_{ik}^{\text{TT}}) + \frac{1}{2}(\delta_{il}\partial_j\partial_k\Theta + \delta_{jk}\partial_i\partial_l\Theta - \delta_{jl}\partial_i\partial_k\Theta - \delta_{ik}\partial_j\partial_l\Theta). \quad (2.15c)$$

As expected, the Riemann tensor is expressed as the six gauge invariant quantities. The six gauge invariant metric components should not be confused with six propagating degrees of freedom. In linearized Einstein gravity, the scalar and vector sectors are constrained, and only the two tensor polarizations propagate in vacuum.

The SVT decomposition and the above curvature identities are kinematical and do not rely on a particular gravitational action. They therefore apply to metric perturbations about Minkowski spacetime in any diffeomorphism invariant metric theory. The equations of motion, constraint structure, and number of propagating degrees of freedom are, however, theory dependent. In TEGR, the metric decomposition must also be supplemented by the tetrad and inertial spin connection sectors associated with local Lorentz invariance [38].

3 Reduction of topological terms

In this section, we reduce the gravitational Pontryagin and Nieh-Yan four-forms to a constant time spacelike hypersurface V . Working to quadratic order in perturbations about Minkowski spacetime, we use the SVT decomposition to express their hypersurface descendants in terms of the gauge invariant variables Φ , Θ , Ξ_i , and h_{ij}^{TT} . In both cases, the bulk contribution separates into a transverse vector sector and a transverse traceless tensor sector, while the scalar and mixed sectors contribute only through codimension-two surface terms. The vector and tensor contributions obtained from the Nieh-Yan descendant have the one-derivative structures of spin-1 and spin-2 gravitomagnetic helicities, whereas the corresponding Pontryagin contributions contain two additional derivatives and have the structures of gravito-current helicities. We also identify all codimension-two terms and state the boundary conditions under which they may consistently be neglected.

For the Nieh-Yan reduction, we employ the covariant formulation of TEGR, in which the tetrad and the flat inertial spin connection are treated as a Lorentz covariant pair. This makes it possible to separate the physical SVT perturbations from infinitesimal diffeomorphisms and local Lorentz transformations. For the Pontryagin tensor sector, we retain the complete off-shell hypersurface descendant and carefully distinguish it from the kinetic, derivative-weighted helicity that can be introduced on-shell. Their relation involves an additional quantity Q^T and therefore depends on both the initial data and the flux through the codimension-two boundary.

3.1 Pontryagin term

We choose a constant time slice V with future directed unit normal $n_\mu = (-1, 0, 0, 0)$. The four-dimensional orientation is chosen as $\epsilon^{0123} = -1$. Accordingly, the induced Levi-Civita symbol on V

is defined by

$$\epsilon^{ijk} \equiv n_\mu \epsilon^{\mu ijk}, \quad \epsilon^{123} = +1.$$

Let $\Gamma_{\mu\beta}^\alpha dx^\mu$ denote the Levi-Civita connection, the gravitational Chern-Simons term on V is

$$H_{CS} = \int_V d^3x n_\mu \epsilon^{\mu\nu\rho\sigma} \left(\Gamma_{\nu\beta}^\alpha \partial_\rho \Gamma_{\sigma\alpha}^\beta + \frac{2}{3} \Gamma_{\nu\beta}^\alpha \Gamma_{\rho\gamma}^\beta \Gamma_{\sigma\alpha}^\gamma \right). \quad (3.1)$$

It is the hypersurface descendant of the Pontryagin four-form,

$$P_4 \equiv \text{Tr}(\mathbf{R} \wedge \mathbf{R}) = dQ_3(\mathbf{\Gamma}), \quad \mathbf{\Gamma}^\alpha_\beta = \Gamma_{\mu\beta}^\alpha dx^\mu$$

where

$$Q_3(\mathbf{\Gamma}) = \text{Tr} \left(\mathbf{\Gamma} \wedge d\mathbf{\Gamma} + \frac{2}{3} \mathbf{\Gamma} \wedge \mathbf{\Gamma} \wedge \mathbf{\Gamma} \right).$$

and the curvature two-form is

$$\mathbf{R} = d\mathbf{\Gamma} + \mathbf{\Gamma} \wedge \mathbf{\Gamma}.$$

Although the connection is retained only to linear order in $h_{\mu\nu}$, the leading non-vanishing Chern-Simons term is quadratic in the perturbation. Therefore,

$$H_{CS}^{(2)} = \int_V d^3x \epsilon^{ijk} \Gamma_{i\beta}^{(1)\alpha} \partial_j \Gamma_{k\alpha}^{(1)\beta}, \quad (3.2)$$

while the cubic connection term contributes only at $\mathcal{O}(h^3)$.

Assuming boundary conditions that remove the zero modes of the spatial SVT decomposition, a convenient gauge invariant representative of the metric perturbation is

$$[h_{00}] = -2\Phi, \quad [h_{0i}] = \Xi_i, \quad [h_{ij}] = h_{ij}^{\text{TT}} + \Theta \delta_{ij} \quad (3.3)$$

where each term is gauge invariant. A general metric perturbation can then be written as

$$h_{\mu\nu} = [h_{\mu\nu}] + 2\partial_{(\mu} \zeta_{\nu)} \quad (3.4)$$

where

$$\zeta_0 = \gamma - \frac{1}{2} \dot{\lambda}, \quad \zeta_i = \frac{1}{2} (\epsilon_i + \partial_i \lambda). \quad (3.5)$$

It is important that the Pontryagin four-form is gauge invariant at this order, whereas the Chern-Simons three-form is invariant only up to an exact form. Indeed,

$$\Gamma_{\mu\beta}^{(1)\alpha} [h] = \Gamma_{\mu\beta}^{(1)\alpha} [[h]] + \partial_\mu \partial_\beta \zeta^\alpha. \quad (3.6)$$

Substitution into (3.2) gives

$$H_{CS}^{(2)}[h; V] = H_{CS}^{(2)}[[h]; V] + \int_{\partial V} d^2y \sqrt{q} r_i \mathcal{J}_\zeta^i, \quad (3.7)$$

where

$$\mathcal{J}_\zeta^i = \epsilon^{ijk} (\partial_\beta \zeta^\alpha) \partial_j \Gamma_{k\alpha}^{(1)\beta} [[h]]. \quad (3.8)$$

Consequently, the bulk part of the hypersurface descendant is fixed by the gauge invariant SVT modes, but the Chern-Simons term on a hypersurface with boundary retains a codimension-two surface ambiguity. This term vanishes when $\partial V = \emptyset$, when the relevant boundary conditions are imposed, or when the gauge invariant representative $\zeta_\mu = 0$ is chosen. Here the coordinates y^A , $A = 1, 2$, are intrinsic coordinates on ∂V , r_k is the outward-pointing unit normal to ∂V within V , and

$$q = \det(q_{AB})$$

is the determinant of the induced two-dimensional metric. Since the integrand is already quadratic in the perturbations, r_k and q_{AB} may be evaluated using the background geometry at this order.

Substituting the gauge invariant SVT representative $[h_{\mu\nu}]$ into (3.2) and organizing the result according to its spatial $SO(3)$ content, we find that only the vector-vector and tensor-tensor sectors contribute to the bulk integral. The pure scalar sector gives no bulk contribution, whereas the non-vanishing terms involving Θ , as well as the mixed vector-tensor terms, can be written as total spatial divergences. After performing the spatial integrations by parts according to the convention adopted below, the quadratic Chern-Simons term evaluated on the gauge invariant representative takes the form

$$H_{\text{CS}}^{(2)}[[h]] = H_{\text{CS}}^V + H_{\text{CS},0}^T + B_{\text{CS}}^{\text{SVT}}. \quad (3.9)$$

The vector and tensor bulk contributions are

$$H_{\text{CS}}^V = \frac{1}{2} \int d^3x \epsilon^{ijk} \partial_j \Xi_i \nabla^2 \Xi_k, \quad (3.10a)$$

$$H_{\text{CS},0}^T = \frac{1}{2} \int d^3x \epsilon^{ijk} \left[\dot{h}_{i\ell}^{\text{TT}} \partial_j \dot{h}_{k\ell}^{\text{TT}} + h_{i\ell}^{\text{TT}} \partial_j \nabla^2 h_{k\ell}^{\text{TT}} \right]. \quad (3.10b)$$

We add a subscript "0" to the tensor sector for now. We add a subscript "0" to the tensor sector for now. The notation H_{CS}^T introduced below is reserved for an improved on-shell kinetic helicity. The term $B_{\text{CS}}^{\text{SVT}}$ is supported on the codimension-two boundary ∂V

$$B_{\text{CS}}^{\text{SVT}} = \int_{\partial V} d^2y \sqrt{q} r_k \mathcal{B}_{\text{SVT}}^k, \quad (3.11)$$

where

$$\mathcal{B}_{\text{SVT}}^k = \mathcal{B}_{\Xi\Xi}^k + \mathcal{B}_{hh}^k + \mathcal{B}_{\Theta\Xi}^k + \mathcal{B}_{h\Xi}^k + \mathcal{B}_{\Theta h}^k. \quad (3.12)$$

The explicit expressions for the five contributions in (3.12) will be given below. For a general representative with $\zeta_\mu \neq 0$, the Chern-Simons term contains an additional gauge-dependent codimension-two term:

$$H_{\text{CS}}^{(2)}[h] = H_{\text{CS}}^V + H_{\text{CS},0}^T + B_{\text{CS}}^{\text{SVT}} + B_\zeta, \quad (3.13)$$

where

$$B_\zeta = \int_{\partial V} d^2y \sqrt{q} r_i \mathcal{J}_\zeta^i. \quad (3.14)$$

Thus, the bulk terms are completely determined by the gauge invariant SVT modes, while the dependence on the choice of representative is confined to ∂V . We now discuss the vector bulk contribution, the tensor bulk contribution, and the codimension-two boundary terms separately.

Spin-1 sector. Since the gauge invariant vector mode is transverse,

$$\nabla \cdot \Xi = 0,$$

we define its associated gravitomagnetic field and gravito-current by

$$\mathbf{B}_\Xi \equiv \nabla \times \Xi, \quad \mathbf{J}_\Xi \equiv \nabla \times \mathbf{B}_\Xi. \quad (3.15)$$

Transversality implies

$$\mathbf{J}_\Xi = \nabla \times (\nabla \times \Xi) = -\nabla^2 \Xi. \quad (3.16)$$

Equation (3.10a) can therefore be written as

$$H_{\text{CS}}^V = \frac{1}{2} \int_V d^3x \mathbf{B}_\Xi \cdot \mathbf{J}_\Xi = -\frac{1}{2} \int_V d^3x (\nabla \times \Xi) \cdot \nabla^2 \Xi. \quad (3.17)$$

Thus, H_{CS}^V has the algebraic structure of a current helicity [39] rather than that of a magnetic helicity. In magnetohydrodynamics, the magnetic and current helicity densities are, respectively,

$$h_{\text{m}} = \mathbf{a} \cdot \mathbf{b}, \quad \mathbf{b} = \nabla \times \mathbf{a}, \quad (3.18a)$$

$$h_{\text{c}} = \mathbf{b} \cdot (\nabla \times \mathbf{b}). \quad (3.18b)$$

Consequently, the correspondence relevant here is

$$\mathbf{b} \longleftrightarrow \mathbf{B}_{\Xi}, \quad \nabla \times \mathbf{b} \longleftrightarrow \mathbf{J}_{\Xi}.$$

This motivates referring to H_{CS}^V as the spin-1 gravito-current helicity. By contrast, the gravitomagnetic helicity discussed in the literature has the magnetic helicity structure

$$\text{gravitomagnetic helicity} \propto \int_V d^3x \, \Xi \cdot (\nabla \times \Xi) \quad (3.19)$$

up to normalization and sign conventions [12]. In the conventions adopted in the present work, this structure appears in the spin-1 Nieh-Yan descendant,

$$H_{\text{NY}}^V = -\frac{1}{4} \int_V d^3x \, \Xi \cdot \mathbf{B}_{\Xi}. \quad (3.20)$$

The Pontryagin expression (3.17) contains two additional spatial derivatives relative to this spin-1 gravitomagnetic helicity.

The gauge invariant bulk density selected by (3.10a) is

$$\mathcal{H}_{\text{CS}}^V = \frac{1}{2} \mathbf{B}_{\Xi} \cdot \mathbf{J}_{\Xi} = -\frac{1}{2} (\nabla \times \Xi) \cdot \nabla^2 \Xi. \quad (3.21)$$

Because Ξ_i is invariant under linearized diffeomorphisms, this bulk representative is also gauge invariant at the linearized level. Nevertheless, $\mathcal{H}_{\text{CS}}^V$ is a density defined on the chosen spatial foliation rather than a four-dimensional scalar. Moreover, as usual for a hypersurface descendant, it is defined only up to a total spatial divergence, whose integral contributes on ∂V . The expression (3.21) is also related algebraically to the magnetic part of Lipkin's electromagnetic zilch. In one commonly used normalization, the vacuum electromagnetic zilch density is [17–19]

$$\rho_{\text{zilch}} = \frac{1}{2} [\mathbf{e} \cdot (\nabla \times \mathbf{e}) + \mathbf{b} \cdot (\nabla \times \mathbf{b})]. \quad (3.22)$$

Here $\mathbf{e}$ denotes the electric field. Our spin-1 density corresponds in form only to the second, magnetic contribution under the replacement

$$\mathbf{b} \longleftrightarrow \mathbf{B}_{\Xi}.$$

It should therefore not be identified with one half of the complete zilch density in general. Such an identification is possible only under additional conditions, for example after an averaging procedure for which the electric and magnetic contributions are equal. The analogy is algebraic, since the spin-1 gravitational mode obeys a constraint equation rather than the source-free Maxwell evolution equations. For linearized Einstein gravity, the vector constraint in the normalization used here is

$$\nabla^2 \Xi_i = -16\pi G T_{0i}^{\text{T}}. \quad (3.23)$$

It follows that

$$\mathcal{H}_{\text{CS}}^V = 8\pi G \mathbf{B}_{\Xi} \cdot \mathbf{T}_0^{\text{T}}, \quad (\mathbf{T}_0^{\text{T}})_i = T_{0i}^{\text{T}}. \quad (3.24)$$

Hence the bulk density vanishes pointwise in every smooth source-free region:

$$T_{0i}^{\text{T}} = 0 \implies \mathcal{H}_{\text{CS}}^V = 0. \quad (3.25)$$

Here "source-free" means $T_{0i}^T = 0$, rather than merely $T_{0i} = 0$ on an open set. Because the transverse projector is spatially nonlocal, $T_{0i} = 0$ outside the support of the physical stress tensor does not in general imply $T_{0i}^T = 0$ there. The pointwise implication above for $\mathcal{H}_{\text{CS}}^V$ therefore applies only after the projected source has been shown to vanish. In a modified theory of gravity, the vector constraint must be derived from the corresponding field equations. If it permits $\nabla^2 \Xi_i \neq 0$ in a region without ordinary matter, then $\mathcal{H}_{\text{CS}}^V$ can be nonvanishing there.

Spin-2 sector. For any symmetric spatial tensor t_{ij} , we define the symmetrized curl operator

$$(Ct)_{ij} \equiv \epsilon_{kl(i} \partial^k t_{j)}^l = \frac{1}{2} (\epsilon_{kli} \partial^k t_j^l + \epsilon_{klj} \partial^k t_i^l). \quad (3.26)$$

The operator C maps a transverse traceless tensor into another transverse traceless tensor and commutes with ∂_t and ∇^2 . Equation (3.10b) becomes

$$H_{\text{CS},0}^T = \frac{1}{2} \int_V d^3x \left[\dot{h}_{ij}^{\text{TT}} (C\dot{h}^{\text{TT}})_{ij} + h_{ij}^{\text{TT}} \nabla^2 (Ch^{\text{TT}})_{ij} \right]. \quad (3.27)$$

This is the complete quadratic spin-2 contribution to the Pontryagin Chern-Simons hypersurface descendant. No field equation has been used in deriving (3.27). The result is therefore kinematical within linear perturbation theory around Minkowski spacetime. Closely related parity odd tensor structures occur in cosmological calculations of the gravitational Pontryagin density [20, 21, 40], although their normalizations, backgrounds, and treatments of total derivatives need not coincide with those used here.

In a source-free region of linearized Einstein gravity, the tensor mode satisfies

$$\ddot{h}_{ij}^{\text{TT}} - \nabla^2 h_{ij}^{\text{TT}} = 0. \quad (3.28)$$

This equation is not universal in modified theories of gravity. On solutions of (3.28), define

$$Q^T = \frac{1}{2} \int_V d^3x \left[\dot{h}_{ij}^{\text{TT}} (C\dot{h}^{\text{TT}})_{ij} - h_{ij}^{\text{TT}} \nabla^2 (Ch^{\text{TT}})_{ij} \right]. \quad (3.29)$$

We promote Q^T to a 3-form $\mathcal{J}_Q^T$:

$$\begin{aligned} \mathcal{J}_Q^T &= \frac{1}{2} \left[\dot{h}_{ij}^{\text{TT}} (Ch^{\text{TT}})^{ij} - h_{ij}^{\text{TT}} \nabla^2 (Ch^{\text{TT}})^{ij} \right] d^3x \\ &+ \frac{1}{4} dt \wedge \epsilon_{klm} \left[(\partial^k h_{ij}^{\text{TT}}) (Ch^{\text{TT}})^{ij} - h_{ij}^{\text{TT}} \partial^k (Ch^{\text{TT}})^{ij} \right] dx^\ell \wedge dx^m. \end{aligned} \quad (3.30)$$

The first term corresponds to the quantity Q^T , while the second term vanishes upon pullback to a spacelike hypersurface. Furthermore, $\mathcal{J}_Q^T$ is closed under exterior differentiation,

$$d\mathcal{J}_Q^T \approx 0, \quad (3.31)$$

where $\approx$ denotes equality on-shell (i.e., after imposing the equations of motion). Hence, we can define a new 3-form Q_3^{new} as

$$Q_3^{\text{new}} = Q_3 + \mathcal{J}_Q^T, \quad (3.32)$$

such that

$$dQ_3^{\text{new}} \approx P_4. \quad (3.33)$$

Therefore, pulling back Q_3^{new} to a spacelike hypersurface yields the density $\mathcal{H}_{\text{CS,kin}}^T$,

$$\mathcal{H}_{\text{CS,kin}}^T = \dot{h}_{ij}^{\text{TT}} (C\dot{h}^{\text{TT}})_{ij}. \quad (3.34)$$

For a fixed coordinate region V , Eq. (3.28) gives

$$\frac{dQ^T}{dt} = \frac{1}{2} \int_{\partial V} d^2y \sqrt{q} r_k \left[(\partial^k h_{ij}^{\text{TT}}) (C \dot{h}^{\text{TT}})_{ij} - h_{ij}^{\text{TT}} \partial^k (C \dot{h}^{\text{TT}})_{ij} \right]. \quad (3.35)$$

Thus Q^T is conserved only if this flux vanishes—for instance, on a boundaryless slice, for fields supported away from the boundary, or under suitable boundary/falloff conditions that make the displayed integral zero.

In the preceding derivation of $\mathcal{H}_{\text{CS,kin}}^T$, the pullback is taken to the constant-time spacelike hypersurface V . Under this pullback, the term containing dt vanishes. Note that the presence of Q^T modifies the bulk density.

Combining (3.27) and (3.29), we obtain the exact identity

$$H_{\text{CS},0}^T = H_{\text{CS,kin}}^T - Q^T, \quad H_{\text{CS,kin}}^T \equiv \int_{V_t} d^3x \dot{h}_{ij}^{\text{TT}} (C \dot{h}^{\text{TT}})_{ij}. \quad (3.36)$$

The corresponding bulk densities (3.27) and (3.34) should be distinguished. This subtraction is an on-shell improvement of the complete descendant, not an identification of the two densities. The complete spin-2 Pontryagin hypersurface descendant remains the density (3.27), while $\dot{h}_{ij}^{\text{TT}} (C \dot{h}^{\text{TT}})_{ij}$ is a physically distinct on-shell kinetic helicity. The latter is not the direct spin-2 Pontryagin descendant. As we shall justify in the next section, the second density provides a more appropriate description of helicity than the first. Here "more appropriate" refers to the helicity observable, not to the topological descent. The descent itself selects (3.27). The kinetic density is obtained only after using the wave equation and subtracting a conserved improvement term. We therefore omit the subscript "kin" and introduce the spin-2 gravito-current helicity

$$H_{\text{CS}}^T = \int d^3x \dot{h}_{ij}^{\text{TT}} (C \dot{h}^{\text{TT}})_{ij}, \quad (3.37)$$

whose density reads

$$\mathcal{H}_{\text{CS}}^T = \dot{h}_{ij}^{\text{TT}} (C \dot{h}^{\text{TT}})_{ij}. \quad (3.38)$$

Boundary term. For the gauge invariant representative $\zeta_\mu = 0$, the codimension-two contribution in (3.9) is

$$\mathcal{B}_{\text{CS}}^{\text{SVT}} = \int_{\partial V} d^2y \sqrt{q} r_m \mathcal{B}_{\text{SVT}}^m, \quad \mathcal{B}_{\text{SVT}}^m = \mathcal{B}_{\Xi\Xi}^m + \mathcal{B}_{hh}^m + \mathcal{B}_{\Theta\Xi}^m + \mathcal{B}_{h\Xi}^m + \mathcal{B}_{\Theta h}^m, \quad (3.39)$$

where

$$\mathcal{B}_{\Xi\Xi}^m = \frac{1}{2} \epsilon^{ijk} \Xi_i \left(\partial^m \partial_j \Xi_k - \delta_j^m \nabla^2 \Xi_k \right), \quad (3.40a)$$

$$\mathcal{B}_{hh}^m = \frac{1}{2} \epsilon^{ijk} h_{i\ell}^{\text{TT}} \left(\partial_j \partial_\ell h_{mk}^{\text{TT}} - \partial_j \partial^m h_{k\ell}^{\text{TT}} \right), \quad (3.40b)$$

$$\mathcal{B}_{\Theta\Xi}^m = -\frac{1}{2} \epsilon^{mij} \Xi_i \partial_j \dot{\Theta}, \quad (3.40c)$$

$$\mathcal{B}_{h\Xi}^m = -\frac{1}{2} \epsilon^{ijk} \left(\dot{h}^{\text{TT}}_{i^m} \partial_j \Xi_k + \Xi_i \partial_j \dot{h}^{\text{TT}}_{k^m} \right), \quad (3.40d)$$

$$\mathcal{B}_{\Theta h}^m = \frac{1}{2} \epsilon^{ijk} (\partial_j \Theta) (\partial_k h^{\text{TT}}_{i^m}). \quad (3.40e)$$

Here m is a free spatial index, whereas r_m denotes the outward-pointing unit normal to ∂V . The first two terms arise from the vector-vector and tensor-tensor sectors, respectively. The remaining terms

originate from the scalar-vector, vector-tensor, and scalar-tensor sectors. For a general representative with $\zeta_\mu \neq 0$, the complete surface contribution is instead

$$B_{\text{CS}}^{\text{total}} = B_{\text{CS}}^{\text{SVT}} + B_\zeta, \quad B_\zeta = \int_{\partial V_t} d^2y \sqrt{q} r_m \mathcal{J}_\zeta^m, \quad (3.41)$$

where $\mathcal{J}_\zeta^m$ is given in (3.8). Whether these terms vanish depends on the hypersurface, the asymptotic region, and the boundary conditions. We have discussed the conditions for a weak field, slowly moving isolated binary in its center-of-mass frame in Appendix A.

In the following, we focus on the local gauge invariant bulk representatives $\mathcal{H}_{\text{CS}}^V$ and $\mathcal{H}_{\text{CS}}^T$. This is a restriction of scope, rather than a consequence of gauge invariance. Whenever the complete integrated Chern-Simons descendant is considered for a finite region or at null infinity, the surface terms must be restored explicitly.

3.2 Nieh-Yan term

The Nieh-Yan form is naturally defined in Riemann-Cartan geometry, where the coframe and the metric are related by

$$g_{\mu\nu} = \eta_{ab} e^a{}_\mu e^b{}_\nu, \quad e^a = e^a{}_\mu dx^\mu. \quad (3.42)$$

For a metric-compatible spin connection, $\omega_{ab} = -\omega_{ba}$, the torsion and curvature two-forms are

$$T^a = de^a + \omega^a{}_b \wedge e^b, \quad R^a{}_b = d\omega^a{}_b + \omega^a{}_c \wedge \omega^c{}_b. \quad (3.43)$$

The Nieh-Yan four-form is the exact form [30]

$$N_{\text{NY}} \equiv T^a \wedge T_a - e^a \wedge e^b \wedge R_{ab} = d(e^a \wedge T_a). \quad (3.44)$$

Exactness alone does not imply that its integral is a nontrivial or quantized topological invariant. If $e^a \wedge T_a$ is globally defined and smooth on a compact manifold without boundary, the integral of N_{NY} vanishes. Nonzero values can instead depend on boundaries, singularities, torsional defects, or global properties of the coframe. This qualification is also important in applications to the chiral anomaly, where the coefficient of the Nieh-Yan contribution can depend on the regulator scale [33, 34].

For the torsion-free Levi-Civita connection used in metric general relativity, $T^a = 0$, and hence $N_{\text{NY}} = 0$. Teleparallel gravity provides a different geometric representation of the same Einstein dynamics where the teleparallel spin connection is flat, while its torsion is generally nonzero. The historical development of teleparallel gravity can be traced to Einstein's work on distant parallelism [41] and to subsequent tetrad and translational gauge formulations [42–44]. Modern reviews include [38, 45, 46]. Recent work has also used the teleparallel Nieh-Yan boundary form to construct a spin-2 helicity flux at $\mathcal{I}^+$ [25]. The quantity considered below is instead the descendant on a spacelike constant time hypersurface.

In TEGR,

$$R^a{}_b = 0, \quad N_{\text{NY}} = T^a \wedge T_a = d(e^a \wedge T_a). \quad (3.45)$$

For a spacetime M bounded by two constant time hypersurfaces V_{t_1} and V_{t_2} , together with an outer boundary $\mathcal{B}$, Stokes' theorem gives

$$\int_M N_{\text{NY}} = H_{\text{NY}}[V_{t_2}] - H_{\text{NY}}[V_{t_1}] + \int_{\mathcal{B}} e^a \wedge T_a, \quad H_{\text{NY}}[V] \equiv \int_V e^a \wedge T_a, \quad (3.46)$$

where every three-form is understood to be pulled back to the corresponding boundary hypersurface.

At linear order, a coframe reproducing the SVT decomposition of the metric can be written as

$$e^{\hat{0}}_0 = 1 - \phi, \quad (3.47a)$$

$$e^{\hat{0}}_i = -\frac{1}{2}(\beta_i + \partial_i \gamma) - \mathbf{b}_i, \quad (3.47b)$$

$$e^{\hat{i}}_0 = \delta^{\hat{i}i} \left[\frac{1}{2}(\beta_i + \partial_i \gamma) - \mathbf{b}_i \right], \quad (3.47c)$$

$$e^{\hat{i}}_j = \delta^{\hat{i}i}_j + \delta^{\hat{i}i} \left[\frac{1}{2} h^{\text{TT}}_{ij} + \frac{1}{6} H \delta_{ij} + \frac{1}{2} \partial_{(i} \epsilon_{j)} + \frac{1}{2} \left(\partial_i \partial_j - \frac{1}{3} \delta_{ij} \nabla^2 \right) \lambda + \epsilon_{ij}{}^k \mathbf{r}_k \right]. \quad (3.47d)$$

Here $\mathbf{b}_i$ and $\mathbf{r}_i$ parameterize the independent infinitesimal Lorentz transformation of the coframe

$$\ell_{\hat{0}\hat{i}} = \mathbf{b}_i, \quad \ell_{\hat{i}\hat{j}} = \delta_{\hat{i}i} \delta_{\hat{j}j} \epsilon^{ijk} \mathbf{r}_k, \quad \ell_{ab} = -\ell_{ba}. \quad (3.48)$$

Let

$$\bar{e}^{\hat{0}} = dt, \quad \bar{e}^{\hat{i}} = \delta^{\hat{i}i}_i dx^i \quad (3.49)$$

be the background coframe. A general linearized coframe can be organized as

$$e^a = [e^a] + d\zeta^a + \Lambda^a{}_b \bar{e}^b + O(2), \quad \Lambda_{ab} = \ell_{ab} + \partial_{[a} \zeta_{b]}. \quad (3.50)$$

Here

$$\zeta_0 = \gamma - \frac{1}{2} \dot{\lambda}, \quad \zeta_i = \frac{1}{2} (\epsilon_i + \partial_i \lambda), \quad \zeta^a = \bar{e}^a{}_\mu \zeta^\mu. \quad (3.51)$$

The quantity Λ_{ab} is the total antisymmetric part that remains after the diffeomorphism contribution has been written as $d\zeta^a$. The representative constructed from the gauge invariant SVT modes is

$$[e^{\hat{0}}] = (1 + \Phi) dt - \frac{1}{2} \Xi_i dx^i, \quad (3.52a)$$

$$[e^{\hat{i}}] = \frac{1}{2} \Xi^i dt + \left[\left(1 + \frac{1}{2} \Theta \right) \delta^{\hat{i}}_j + \frac{1}{2} h^{\text{TT}\hat{i}}{}_j \right] dx^j. \quad (3.52b)$$

Its coefficients are invariant under linearized diffeomorphisms, subject to the boundary conditions used to define the SVT decomposition. The coframe itself is still a Lorentz vector and must be accompanied by the corresponding inertial spin connection. To determine that connection, one switches off the gravitational modes in (3.50) and introduces the reference coframe

$$E^a = \bar{e}^a + d\zeta^a + \Lambda^a{}_b \bar{e}^b + \text{higher orders}. \quad (3.53)$$

The inertial spin connection is required to satisfy

$$dE^a + \omega^a{}_b \wedge E^b = 0, \quad R^a{}_b(\omega) = 0. \quad (3.54)$$

Locally, a flat spin connection can be written as

$$\omega^a{}_b = (L dL^{-1})^a{}_b, \quad L^a{}_b = \delta^a{}_b + \Lambda^a{}_b + \text{higher orders}, \quad (3.55)$$

and hence

$$\omega^a{}_b = -d\Lambda^a{}_b + \text{higher orders}. \quad (3.56)$$

This is the covariant reference-coframe prescription of TEGR [38, 47]. Flatness implies that the connection is pure gauge locally. A global Weitzenböck gauge can be obstructed by nontrivial holonomy. It is important that the symmetric coframe choice $\ell_{ab} = 0$ does not generally imply $\omega^a{}_b = 0$, because then $\Lambda_{ab} = \partial_{[a} \zeta_{b]}$. The Weitzenböck gauge instead requires

$$\Lambda_{ab} = 0 \quad \Longleftrightarrow \quad \ell_{ab} = -\partial_{[a} \zeta_{b]}. \quad (3.57)$$

The simultaneous choice $\zeta^a = 0$ and $\ell_{ab} = 0$ is a particularly simple representative for which both $\Lambda_{ab} = 0$ and $\omega^a_b = 0$. Define the gauge invariant coframe perturbation by

$$\delta e_{\text{GI}}^a \equiv [e^a] - \bar{e}^a. \quad (3.58)$$

Using (3.50) and (3.56), the linearized torsion is

$$T^{a(1)} = d\delta e_{\text{GI}}^a. \quad (3.59)$$

The terms containing $d\Lambda^a_b$ cancel between de^a and $\omega^a_b \wedge e^b$, while $d^2\zeta^a = 0$. More generally, under a simultaneous Lorentz transformation of the coframe and spin connection,

$$e^a \longrightarrow L^a_b e^b, \quad T^a \longrightarrow L^a_b T^b, \quad (3.60)$$

so that $e^a \wedge T_a$ is Lorentz invariant. Because of this Lorentz invariance, the quadratic descendant may be evaluated locally in the Weitzenbock gauge $\Lambda_{ab} = 0$. In this gauge,

$$e^a = [e^a] + d\zeta^a + \text{higher orders}, \quad \omega^a_b = 0, \quad T^{a(1)} = d\delta e_{\text{GI}}^a. \quad (3.61)$$

Evaluating the same expression in an arbitrary Lorentz frame gives the same result when the coframe and inertial spin connection are retained consistently to the required perturbative order.

Keeping the quadratic functional constructed from the first-order perturbations, the pullback of the Nieh-Yan boundary form to the hypersurface V becomes

$$\begin{aligned} H_{\text{NY}}^{(2)}[V] &= \int_V \delta e_{\text{GI}}^a \wedge d\delta e_{\text{GI},a} + \int_V d\zeta^a \wedge d\delta e_{\text{GI},a} \\ &= \int_V \delta e_{\text{GI}}^a \wedge d\delta e_{\text{GI},a} + \int_{\partial V} \zeta^a d\delta e_{\text{GI},a} \\ &= H_{\text{NY}}^V + H_{\text{NY}}^T + B_{\text{NY}}^\zeta. \end{aligned} \quad (3.62a)$$

All Lorentz-frame dependence has canceled through the inertial spin connection. The remaining surface term records the use of a general diffeomorphism representative on a hypersurface with boundary. Neither scalar mode contributes to the bulk expression. The codimension-two term is

$$B_{\text{NY}}^\zeta = \int_{\partial V} d^2y \sqrt{q} r_m \mathcal{J}_{\text{NY}}^m, \quad (3.63)$$

where

$$\mathcal{J}_{\text{NY}}^m = \frac{1}{2} \epsilon^{mjk} \left[\zeta^{\hat{0}} \partial_j \Xi_k + \zeta^{\hat{\ell}} \partial_j (\Theta \delta_{\ell k} + h_{\ell k}^{\text{TT}}) \right]. \quad (3.64)$$

The condition $\zeta^a|_{\partial V} = 0$ is sufficient, but not necessary, for B_{NY}^ζ to vanish. Transformations that do not vanish at the boundary can change the fixed-hypersurface representative by this surface term, which must then be retained. The corresponding bulk densities are

$$\mathcal{H}_{\text{NY}}^V = -\frac{1}{4} \Xi \cdot (\nabla \times \Xi), \quad (3.65a)$$

$$\mathcal{H}_{\text{NY}}^T = \frac{1}{4} h_{ij}^{\text{TT}} (Ch^{\text{TT}})_{ij}. \quad (3.65b)$$

The first expression has the structure of gravitomagnetic helicity [12]. Unlike the electromagnetic vector potential, Ξ_i is invariant under linearized diffeomorphisms. The second expression is a spin-2 curl pairing closely related to duality-based helicity constructions for linearized gravity [13–15]. Both are gauge invariant bulk representatives on the chosen spatial foliation. They are not four-dimensional scalar densities and remain defined modulo total spatial divergences.

Topological origin	Expression	Terminology
Nieh-Yan	$H_{NY}^V = -\frac{1}{4} \int d^3x \epsilon_{ijk} \Xi_i \partial_j \Xi_k$	spin-1 gravitomagnetic helicity
Nieh-Yan	$H_{NY}^T = \frac{1}{4} \int d^3x h_{ij}^{\text{TT}} C h_{ij}^{\text{TT}}$	spin-2 gravitomagnetic helicity
Pontryagin	$H_{CS}^V = -\frac{1}{2} \int d^3x \epsilon_{ijk} \partial_j \Xi_k \nabla^2 \Xi_i$	spin-1 gravito-current helicity
Pontryagin	$H_{CS}^T = \int d^3x \dot{h}_{ij}^{\text{TT}} C \dot{h}_{ij}^{\text{TT}}$	spin-2 gravito-current helicity

Table 1: The first three rows are sectors of quadratic spacelike hypersurface descendants, modulo codimension-two surface terms. The final row is a separately defined on-shell kinetic functional related to the complete spin-2 Pontryagin descendant by Eq. (3.36).

3.3 Summary

Table 1 summarizes the four parity odd, gauge invariant bulk sectors that arise in the quadratic reduction of the Nieh-Yan and Pontryagin forms to a constant time hypersurface V . They should be distinguished from the complete hypersurface descendants, which also contain codimension-two contributions. The Nieh-Yan bulk functionals contain one derivative, whereas the Pontryagin bulk functionals contain three derivatives. Motivated by the magnetic and current helicity terminology, we refer to the former as spin- s gravitomagnetic helicities and to the latter as spin- s gravito-current helicities. For the spin-1 sector, this terminology is consistent with the gravitomagnetic helicity construction of [12]. The spin-2 terminology is a proposal of the present work, motivated by analogous curl pairings in duality-based descriptions of linearized gravity [13–15].

The word "spin" in Table 1 labels the irreducible spatial $SO(3)$ sector of the SVT decomposition. It does not imply that every listed field is an independent propagating degree of freedom. Indeed, in linearized Einstein gravity, Ξ_i is determined by a vector constraint, whereas h_{ij}^{TT} carries the two radiative degrees of freedom. An important qualification concerns the spin-2 Pontryagin sector. With the bulk reduction adopted above, the density selected by the Pontryagin descent is (3.27). This modified spin-2 bulk descendant must be distinguished from the separately defined density (3.38). The latter is algebraically related to the spin-2 gravitomagnetic helicity density by

$$\mathcal{H}_{CS}^T = 4 \mathcal{H}_{NY}^T \Big|_{h_{ij}^{\text{TT}} \rightarrow \dot{h}_{ij}^{\text{TT}}} . \quad (3.66)$$

The factor of four follows from the normalization.

The terminology adopted above admits a representation theoretic interpretation. With Pauli-Lubanski operator

$$W^\mu = -\frac{1}{2} \epsilon^{\mu\nu\rho\sigma} P_\nu M_{\rho\sigma},$$

the temporal component is $W^0 = \mathbf{P} \cdot \mathbf{J}$. On the transverse spin-1 and TT spin-2 sectors, respectively,

$$(W_{(1)}^0 \Xi)_i = (\nabla \times \Xi)_i, \quad (W_{(2)}^0 h^{\text{TT}})_{ij} = 2(C h^{\text{TT}})_{ij}. \quad (3.67)$$

The factor of two in the second relation arises because the rotation generator acts on both indices of a symmetric tensor. Consequently,

$$\mathcal{H}_{NY}^V = -\frac{1}{4} \Xi_i (W_{(1)}^0 \Xi)_i, \quad \mathcal{H}_{NY}^T = \frac{1}{8} h_{ij}^{\text{TT}} (W_{(2)}^0 h^{\text{TT}})_{ij}.$$

The corresponding Pontryagin inspired expressions contain two additional derivatives:

$$\mathcal{H}_{CS}^V = -\frac{1}{2} (W_{(1)}^0 \Xi)_i \nabla^2 \Xi_i, \quad \mathcal{H}_{CS}^T = \frac{1}{2} \dot{h}_{ij}^{\text{TT}} (W_{(2)}^0 \dot{h}^{\text{TT}})_{ij}.$$

Thus the Nieh-Yan quantities measure first moments of the helicity spectrum, whereas the Pontryagin inspired quantities measure higher wave number or frequency moments of the same spectrum. This interpretation should not be taken to mean that the local densities are unique Pauli-Lubanski charge densities. The SVT decomposition is tied to a spacelike foliation, and local densities remain sensitive to total-divergence improvements. Moreover, in linearized Einstein gravity the vector mode Ξ_i is constrained rather than an independent massless spin-1 particle. The genuine one-particle Pauli-Lubanski relation $W^\mu = \lambda P^\mu$ applies directly to the TT graviton modes, with $\lambda = \pm 2$. The same approach can be extended to other spinning fields. For more details, please read Appendix B. A representation-theoretic extension to free massless fields of arbitrary integer spin is also discussed in the concluding section.

4 Mode expansion

In Einstein gravity, the spin-1 and spin-2 modes satisfy the following equations

$$\nabla^2 \Xi_i = -16\pi G T_{0i}^T, \quad (4.1a)$$

$$\square h_{ij}^{\text{TT}} = -16\pi G T_{ij}^{\text{TT}} \quad (4.1b)$$

where T_{0i}^T is the transverse part of the momentum density and T_{ij}^{TT} is the transverse and traceless part of the stress tensor T_{ij} . The first equation (4.1a) is a Poisson equation and thus Ξ_i is not the propagating modes. The second equation (4.1b) shows that h_{ij}^{TT} are truly propagating degrees of freedom. The solutions are

$$\Xi_i(t, \mathbf{x}) = 4G \int d^3 \mathbf{x}' \frac{T_{0i}^T(t, \mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} + \Xi_i^{\text{hom}}, \quad (4.2a)$$

$$h_{ij}^{\text{TT}}(t, \mathbf{x}) = 4G \int d^3 \mathbf{x}' \frac{T_{ij}^{\text{TT}}(t - |\mathbf{x} - \mathbf{x}'|, \mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} + h_{ij}^{\text{TT, hom}}. \quad (4.2b)$$

The first solution is instantaneous because the vector equation is a constraint, while the second is retarded. Since the TT projection is spatially nonlocal, (4.2b) should be understood as the retarded Green function solution of the already projected equation. When there is no source in the whole space, the solutions are completely determined by the homogeneous solutions.

The homogeneous solution Ξ_i^{hom} obeys the equation

$$\nabla^2 \Xi_i^{\text{hom}} = 0, \quad \partial_i \Xi_i^{\text{hom}} = 0. \quad (4.3)$$

In the whole space, assuming $\Xi_i^{\text{hom}} \rightarrow 0$ as $r \rightarrow \infty$, the unique solution is $\Xi_i^{\text{hom}} = 0$. We conclude that the spin-1 gravitomagnetic/gravito-current helicities are always zero in the absence of matter

$$\mathcal{H}_{NY}^V = \mathcal{H}_{CS}^V = 0 \quad (4.4)$$

For the spin-2 modes, the homogeneous solution $h_{ij}^{\text{TT, hom}}$ satisfies the equation

$$\square h_{ij}^{\text{TT, hom}} = 0, \quad \partial^i h_{ij}^{\text{TT, hom}} = 0, \quad \delta^{ij} h_{ij}^{\text{TT, hom}} = 0. \quad (4.5)$$

A real solution can be expanded as [48]

$$h_{ij}^{\text{TT, hom}} = \sum_{\sigma=\pm} \int \frac{d^3 \mathbf{k}}{(2\pi)^3} \left[h_\sigma(\mathbf{k}) e^{-i\omega t} + h_\sigma^*(-\mathbf{k}) e^{i\omega t} \right] e_{ij}^\sigma(\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{x}}, \quad \omega = k = |\mathbf{k}| \quad (4.6)$$

where the polarization tensor obeys the condition

$$e_{ij}^{(\sigma)} = e_{ji}^{(\sigma)}, \quad k_i e_{ij}^{(\sigma)} = 0, \quad \delta_{ij} e_{ij}^{(\sigma)} = 0, \quad e_{ij}^{(\sigma)*} e_{ij}^{(\sigma')} = 2\delta_{\sigma\sigma'}, \quad e_{ij}^{(\sigma)*}(\mathbf{k}) = e_{ij}^{(-\sigma)}(\mathbf{k}) = e_{ij}^{(\sigma)}(-\mathbf{k}). \quad (4.7)$$

The completeness relation is

$$\frac{1}{2} \sum_{\sigma=\pm} e_{ij}^{(\sigma)} e_{kl}^{(\sigma)*} = \Pi_{ij,kl}(\hat{k}), \quad (4.8)$$

where

$$\Pi_{ij,kl}(\hat{k}) = \frac{1}{2} \left(P_{ik}(\hat{k}) P_{jl}(\hat{k}) + P_{il}(\hat{k}) P_{jk}(\hat{k}) - P_{ij}(\hat{k}) P_{kl}(\hat{k}) \right), \quad P_{ij}(\hat{k}) = \delta_{ij} - \hat{k}_i \hat{k}_j, \quad \hat{k} = \frac{\mathbf{k}}{k}. \quad (4.9)$$

Note that in the mode expansion, $\sigma \in \{+1, -1\}$ is a chirality label, while the physical spin-2 helicity is given by $s = 2\sigma \in \{+2, -2\}$. Then the spin-2 gravitomagnetic helicity is

$$H_{\text{NY}}^T = \frac{1}{2} \sum_{\sigma=\pm} \sigma \int \frac{d^3 \mathbf{k}}{(2\pi)^3} k |\mathcal{H}_\sigma(t, \mathbf{k})|^2 \quad (4.10)$$

and the gravito-current helicity is

$$H_{\text{CS}}^T = 2 \sum_{\sigma=\pm} \sigma \int \frac{d^3 \mathbf{k}}{(2\pi)^3} k |\dot{\mathcal{H}}_\sigma(t, \mathbf{k})|^2. \quad (4.11)$$

We have defined

$$\mathcal{H}_\sigma(t, \mathbf{k}) = h_\sigma(\mathbf{k}) e^{-i\omega t} + h_\sigma^*(-\mathbf{k}) e^{i\omega t}. \quad (4.12)$$

Therefore,

$$|\mathcal{H}_\sigma|^2 = |h_\sigma(\mathbf{k})|^2 + |h_\sigma(-\mathbf{k})|^2 + 2 \operatorname{Re} [h_\sigma(\mathbf{k}) h_\sigma(-\mathbf{k}) e^{-2i\omega t}], \quad (4.13)$$

$$|\dot{\mathcal{H}}_\sigma|^2 = k^2 \left\{ |h_\sigma(\mathbf{k})|^2 + |h_\sigma(-\mathbf{k})|^2 - 2 \operatorname{Re} [h_\sigma(\mathbf{k}) h_\sigma(-\mathbf{k}) e^{-2i\omega t}] \right\} \quad (4.14)$$

The terms depending on time t reflect the interference effect. The oscillatory terms may disappear after an explicit time average, or through dephasing under suitable assumptions for a continuous spectrum. We define

$$\overline{X} \equiv \lim_{\mathcal{T} \rightarrow \infty} \frac{1}{\mathcal{T}} \int_{t_0}^{t_0 + \mathcal{T}} dt X(t). \quad (4.15)$$

For a monochromatic mode, averaging over an integer number of periods gives the same result. Applying (4.15) to (4.10) and (4.11), and changing $\mathbf{k} \rightarrow -\mathbf{k}$ where needed, yields

$$\overline{H_{\text{NY}}^T} = \int \frac{d^3 \mathbf{k}}{(2\pi)^3} k (|h_+|^2 - |h_-|^2), \quad (4.16a)$$

$$\overline{H_{\text{CS}}^T} = 4 \int \frac{d^3 \mathbf{k}}{(2\pi)^3} k^3 (|h_+|^2 - |h_-|^2). \quad (4.16b)$$

These formulas are signed moments of the classical chiral spectrum. They should not be identified directly with graviton occupation numbers unless the field is canonically normalized and quantized. After canonical normalization of the quadratic Einstein action, $k|h_\sigma|^2$ is proportional, with convention-dependent factors of G , $\hbar$, and the normalization volume, to the occupation number in the helicity- 2σ sector. In that sense $\overline{H_{\text{NY}}^T}$ is proportional to the net chiral occupation, whereas $\overline{H_{\text{CS}}^T}$ weights the same imbalance by two additional powers of frequency. For comparison, mode expansion of the original spin-2 Pontryagin bulk descendant (3.27) is

$$-4 \sum_{\sigma=\pm} \sigma \int \frac{d^3 \mathbf{k}}{(2\pi)^3} k^3 \operatorname{Re} [h_\sigma(\mathbf{k}) h_\sigma(-\mathbf{k}) e^{-2i\omega t}] \quad (4.17)$$

whose time average is zero. Therefore, (3.27) cannot be the candidate quantity of helicity. This conclusion relies on the criterion adopted here for a time-averaged free-field spectral helicity; nevertheless, the vanishing average does not render the complete Pontryagin descendant mathematically trivial.

Electromagnetic analogue. The spectral structure has a useful electromagnetic analogue. The magnetic and current helicity are defined as

$$H_m = \int d^3x \mathbf{a} \cdot \mathbf{b}, \quad (4.18a)$$

$$H_c = \int d^3x \mathbf{b} \cdot (\nabla \times \mathbf{b}). \quad (4.18b)$$

In Coulomb gauge, we use the mode expansion of the magnetic potential

$$a_i(t, \mathbf{x}) = \sum_{s=\pm} \int \frac{d^3\mathbf{k}}{(2\pi)^3} \left[c_s(\mathbf{k}) e^{-ikt} + c_s^*(-\mathbf{k}) e^{ikt} \right] e_i^{(s)}(\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{x}}, \quad k = |\mathbf{k}| \quad (4.19)$$

where the polarization vector $e_i^{(s)}$ satisfies the condition

$$k_i e_i^{(s)} = 0, \quad e_i^{(s)*} e_i^{(s')} = \delta_{ss'}, \quad e_i^{(s)*}(\mathbf{k}) = e_i^{(-s)}(\mathbf{k}), \quad e_i^{(s)}(-\mathbf{k}) = e_i^{(s)*}(\mathbf{k}), \quad (4.20)$$

and the completeness relation

$$\sum_{s=\pm} e_i^{(s)}(\mathbf{k}) e_j^{(s)*}(\mathbf{k}) = P_{ij}(\mathbf{k}) = \delta_{ij} - \hat{k}_i \hat{k}_j. \quad (4.21)$$

Then the magnetic helicity and current helicity become

$$H_m = 2 \sum_{s=\pm} s \int \frac{d^3\mathbf{k}}{(2\pi)^3} k \left[|c_s(\mathbf{k})|^2 + \text{Re}(c_s(\mathbf{k}) c_s(-\mathbf{k}) e^{-2ikt}) \right], \quad (4.22a)$$

$$H_c = 2 \sum_{s=\pm} s \int \frac{d^3\mathbf{k}}{(2\pi)^3} k^3 \left[|c_s(\mathbf{k})|^2 + \text{Re}(c_s(\mathbf{k}) c_s(-\mathbf{k}) e^{-2ikt}) \right]. \quad (4.22b)$$

Average over a period, we find

$$\overline{H_m} = 2 \int \frac{d^3\mathbf{k}}{(2\pi)^3} k \left(|c_+(\mathbf{k})|^2 - |c_-(\mathbf{k})|^2 \right), \quad (4.23a)$$

$$\overline{H_c} = 2 \int \frac{d^3\mathbf{k}}{(2\pi)^3} k^3 \left(|c_+(\mathbf{k})|^2 - |c_-(\mathbf{k})|^2 \right). \quad (4.23b)$$

They have the same spectral form as (4.16), up to normalization, which motivates the terminology of spin-2 gravitomagnetic and gravito-current helicity.

Energy-helicities inequalities. The quadratic Hamiltonian of the free TT field is

$$E = \frac{1}{64\pi G} \int d^3\mathbf{x} \left[\dot{h}_{ij}^{\text{TT}} \dot{h}_{ij}^{\text{TT}} + (\partial_k h_{ij}^{\text{TT}})(\partial_k h_{ij}^{\text{TT}}) \right]. \quad (4.24)$$

Substituting the mode expansion gives

$$E = \frac{1}{32\pi G} \sum_{\sigma=\pm} \int \frac{d^3\mathbf{k}}{(2\pi)^3} \left[\left| \dot{\mathcal{H}}_\sigma(t, \mathbf{k}) \right|^2 + k^2 \left| \mathcal{H}_\sigma(t, \mathbf{k}) \right|^2 \right]$$

$$= \frac{1}{8\pi G} \sum_{\sigma=\pm} \int \frac{d^3\mathbf{k}}{(2\pi)^3} k^2 |h_{\sigma}(\mathbf{k})|^2. \quad (4.25)$$

In the second equality, the oscillatory terms cancel between the kinetic and gradient contributions. Consequently, the total energy is conserved without performing a time average.

Using $d^3\mathbf{k} = k^2 dk d\Omega_{\mathbf{k}}$, we define the angle-integrated energy spectrum by

$$E = \int_0^{\infty} dk \mathcal{E}(k), \quad (4.26)$$

where

$$\mathcal{E}(k) = \frac{k^4}{8\pi G(2\pi)^3} \int d\Omega_{\mathbf{k}} \left[|h_+(k, \Omega_{\mathbf{k}})|^2 + |h_-(k, \Omega_{\mathbf{k}})|^2 \right]. \quad (4.27)$$

Similarly, the time averaged helicities can be written as

$$\overline{H_{\text{NY}}^T} = \int_0^{\infty} dk \mathcal{H}_{\text{NY}}(k), \quad \overline{H_{\text{CS}}^T} = \int_0^{\infty} dk \mathcal{H}_{\text{CS}}(k), \quad (4.28)$$

with

$$\mathcal{H}_{\text{NY}}(k) = \frac{k^3}{(2\pi)^3} \int d\Omega_{\mathbf{k}} \left[|h_+(k, \Omega_{\mathbf{k}})|^2 - |h_-(k, \Omega_{\mathbf{k}})|^2 \right], \quad (4.29)$$

$$\mathcal{H}_{\text{CS}}(k) = \frac{4k^5}{(2\pi)^3} \int d\Omega_{\mathbf{k}} \left[|h_+(k, \Omega_{\mathbf{k}})|^2 - |h_-(k, \Omega_{\mathbf{k}})|^2 \right]. \quad (4.30)$$

In particular,

$$\mathcal{H}_{\text{CS}}(k) = 4k^2 \mathcal{H}_{\text{NY}}(k). \quad (4.31)$$

Since

$$\left| |h_+|^2 - |h_-|^2 \right| \leq |h_+|^2 + |h_-|^2, \quad (4.32)$$

the spectral densities obey the pointwise inequalities

$$|\mathcal{H}_{\text{NY}}(k)| \leq \frac{8\pi G}{k} \mathcal{E}(k), \quad (4.33a)$$

$$|\mathcal{H}_{\text{CS}}(k)| \leq 32\pi G k \mathcal{E}(k). \quad (4.33b)$$

Applying the triangle inequality to the integrals over k , we obtain

$$\left| \overline{H_{\text{NY}}^T} \right| \leq 8\pi G \int_0^{\infty} \frac{dk}{k} \mathcal{E}(k), \quad (4.34a)$$

$$\left| \overline{H_{\text{CS}}^T} \right| \leq 32\pi G \int_0^{\infty} dk k \mathcal{E}(k). \quad (4.34b)$$

Equivalently, writing $dE(k) = \mathcal{E}(k) dk$, these inequalities take the compact form

$$\left| \overline{H_{\text{NY}}^T} \right| \leq 8\pi G \int_0^{\infty} \frac{dE(k)}{k}, \quad (4.35a)$$

$$\left| \overline{H_{\text{CS}}^T} \right| \leq 32\pi G \int_0^{\infty} k dE(k). \quad (4.35b)$$

If the spectrum has support in a finite interval $k_{\min} \leq k \leq k_{\max}$, the preceding relations imply

$$\left| \overline{H_{\text{NY}}^T} \right| \leq \frac{8\pi G}{k_{\min}} E, \quad (4.36a)$$

$$\left| \overline{H_{\text{CS}}^T} \right| \leq 32\pi G k_{\max} E. \quad (4.36b)$$

The first inequality requires a nonzero infrared cutoff, whereas the second requires control of the ultraviolet part of the spectrum. The spectral inequalities are saturated when only one circular polarization is present. Saturation of the integrated inequalities also requires the helicity asymmetry to have the same sign throughout the spectrum. For a monochromatic, purely circularly polarized wave of wave number k_0 , one therefore has

$$\left| \overline{H_{\text{NY}}^T} \right| = \frac{8\pi G}{k_0} E, \quad \left| \overline{H_{\text{CS}}^T} \right| = 32\pi G k_0 E, \quad (4.37)$$

together with

$$\overline{H_{\text{CS}}^T} = 4k_0^2 \overline{H_{\text{NY}}^T}. \quad (4.38)$$

The inequality for H_{CS}^T applies to the spin-2 gravito-current helicity defined in Eq. (4.11). It does not apply without modification to the complete Pontryagin Chern-Simons descendant, which contains the additional $h_{ij}^{\text{TT}} \nabla^2 C h_{ij}^{\text{TT}}$ contribution.

5 Multipole expansion

Suppose that the physical stress energy tensor is spatially localized,

$$T_{\mu\nu}(t, \mathbf{x}') = 0, \quad |\mathbf{x}'| > r_c. \quad (5.1)$$

We impose regularity in the interior and asymptotic decay on the constraint sector, together with the retarded, no-incoming-radiation boundary condition on the tensor sector. These conditions set the independent homogeneous solutions to zero. The resulting formal solutions are

$$\Xi_i(t, \mathbf{x}) = 4G \int d^3 \mathbf{x}' \frac{T_{0i}^{\text{T}}(t, \mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|}, \quad (5.2a)$$

$$h_{ij}^{\text{TT}}(t, \mathbf{x}) = 4G \int d^3 \mathbf{x}' \frac{T_{ij}^{\text{TT}}(t - |\mathbf{x} - \mathbf{x}'|, \mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|}. \quad (5.2b)$$

We have ignored the plane wave solution for the spin-2 modes.

Spin-1 modes. The SVT decomposition separates the momentum density T_{0i} into transverse and longitudinal modes

$$T_{0i} = T_{0i}^{\text{T}} + \partial_i S, \quad \partial^i T_{0i}^{\text{T}} = 0 \quad (5.3)$$

where the transverse mode is

$$T_{0i}^{\text{T}} = \left(\delta_{ij} - \frac{\partial_i \partial_j}{\nabla^2} \right) T_{0j}. \quad (5.4)$$

Here the inverse Laplacian and its zero mode are fixed by the same boundary conditions used in the SVT decomposition. For fields that vanish sufficiently rapidly at spatial infinity, according to (2.13), the projector in (5.4) is consequently nonlocal in position space [35]. Suppose that the stress tensor has support inside a ball of radius r_c . The vector constraint can then be solved either in terms of the projected source or directly in terms of T_{0i}

$$\Xi_i(t, \mathbf{x}) = 4G \int d^3 \mathbf{x}' \frac{T_{0i}^{\text{T}}(t, \mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} = 2G \int d^3 \mathbf{x}' K_{ij}(\mathbf{x} - \mathbf{x}') T_{0j}(t, \mathbf{x}'), \quad (5.5a)$$

where

$$K_{ij}(\mathbf{R}) = \frac{\delta_{ij} + N_i N_j}{R}, \quad \mathbf{R} = \mathbf{x} - \mathbf{x}', \quad R = |\mathbf{R}|, \quad N_i = \frac{R_i}{R}. \quad (5.6)$$

For $r > r_c$, introduce the multipole moments

$$\mathcal{J}_{j|L}(t) = \int d^3\mathbf{x}' x'_{i_1} \cdots x'_{i_\ell} T_{0j}(t, \mathbf{x}'). \quad L = i_1 \cdots i_\ell, \quad (5.7)$$

then we find the multipole expansion in the region $r > r_c$

$$\Xi_i(t, \mathbf{x}) = 2G \sum_{\ell=0}^{\infty} \frac{(-1)^\ell}{\ell!} \mathcal{J}_{j|L}(t) \partial_L K_{ij}(\mathbf{x}). \quad (5.8)$$

The moments $\mathcal{J}_{j|L}$ are symmetric in the multi-index L , but they have not been projected to their STF parts. In contrast to the scalar $1/r$ expansion, their traces cannot in general be discarded because

$$\nabla^2 K_{ij}(\mathbf{x}) = \frac{2}{r^3} (\delta_{ij} - 3n_i n_j), \quad n_i = \frac{x_i}{r}, \quad r \neq 0. \quad (5.9)$$

Thus these moments should not be identified directly with the standard irreducible source multipoles used in gravitational radiation theory. Their decomposition into irreducible Cartesian tensors can be organized by the usual STF methods [49–51]. The first few multipole moments are

$$P_i = \int d^3\mathbf{x}' T_{0i}(t, \mathbf{x}'), \quad (5.10a)$$

$$D_{ij} = \int d^3\mathbf{x}' x'_j T_{0i}(t, \mathbf{x}'). \quad (5.10b)$$

The P_i is the momentum of the system. Using the conservation law of stress tensor, P_i is simplified to

$$P_i = \int d^3\mathbf{x}' \partial'_j (x'_i T_{0j}(t, \mathbf{x}')) - \int d^3\mathbf{x}' x'_i \partial'_j T_{0j}(t, \mathbf{x}') = -\frac{d}{dt} I_i \quad (5.11)$$

where I_i is the dipole moment

$$I_i = \int d^3\mathbf{x}' x'_i T_{00}(t, \mathbf{x}'). \quad (5.12)$$

We have ignored the boundary term by assuming the source is located inside the ball $r < r_c$. Thus the $\ell = 0$ moment contributes to the spin-1 mode as follows

$$\Xi_i^{(\ell=0)}(t, \mathbf{x}) = \frac{2G}{r} (P_i + n_i(\mathbf{P} \cdot \mathbf{n})) = -\frac{2G}{r} (\dot{I}_i + n_i n_j \dot{I}_j). \quad (5.13)$$

In the center-of-mass frame of an isolated system, the momentum is zero and one should consider the next order. The moment with $\ell = 1$ can be decomposed into symmetric traceless, anti-symmetric and trace terms

$$D_{ij} = D_{\langle ij \rangle} + D_{[ij]} + \frac{1}{3} \delta_{ij} D_{kk}. \quad (5.14)$$

For the anti-symmetric part, we define

$$D_{[ij]} = -\frac{1}{2} \epsilon_{ijk} S_k \quad \Rightarrow \quad S_i = \epsilon_{ijk} D_{kj} = \epsilon_{ijk} \int d^3\mathbf{x}' x'_j T_{0k}(t, \mathbf{x}'). \quad (5.15)$$

Therefore, S_i is the total angular momentum of the system. For the symmetric traceless part, we use the conservation of stress tensor

$$D_{\langle ij \rangle} = \int d^3\mathbf{x}' x'_{\langle j} T_{0i \rangle}(t, \mathbf{x}') = -\frac{1}{2} \dot{I}_{ij} \quad (5.16)$$

where I_{ij} is the STF quadrupole moment

$$I_{ij} = \int d^3\mathbf{x}' (x'_i x'_j - \frac{1}{3} \delta_{ij} x'^2) T_{00}(t, \mathbf{x}'). \quad (5.17)$$

Since

$$\partial_k K_{ij} = \frac{1}{r^2} (-3n_i n_j n_k - \delta_{ij} n_k + \delta_{ik} n_j + \delta_{jk} n_i), \quad (5.18)$$

the $\ell = 1$ term is

$$\Xi_i^{(\ell=1)} = \frac{2G}{r^2} (3D_{\langle jk \rangle} n_i n_j n_k + \epsilon_{ijk} S_j n_k). \quad (5.19)$$

The trace term does not contribute to the expansion. Therefore, the leading two orders of the multipole expansion of Ξ_i are

$$\Xi_i = \frac{2G}{r} (P_i + n_i n_j P_j) + \frac{2G}{r^2} (3D_{\langle jk \rangle} n_i n_j n_k + \epsilon_{ijk} S_j n_k) + \mathcal{O}(r^{-3}). \quad (5.20)$$

1. In a general reference of frame, $P_i \neq 0$, the leading terms are

$$\mathcal{H}_{\text{NY}}^V = -\frac{G^2}{r^4} \left[P_i S_i + 3(P_i n_i)(S_j n_j) + 6P_i \epsilon_{ijk} D_{\langle j\ell \rangle} n_\ell n_k \right] + \mathcal{O}(r^{-5}), \quad (5.21a)$$

$$\mathcal{H}_{\text{CS}}^V = \frac{4G^2}{r^6} \left[P_i S_i + 3(P_i n_i)(S_j n_j) + 6P_i \epsilon_{ijk} D_{\langle j\ell \rangle} n_\ell n_k \right] + \mathcal{O}(r^{-7}). \quad (5.21b)$$

Interestingly, the leading order of these two quantities are proportional to each other

$$\mathcal{H}_{\text{CS}}^V = -\frac{4}{r^2} \mathcal{H}_{\text{NY}}^V + \mathcal{O}(r^{-7}). \quad (5.22)$$

This proportionality is not an exact identity between the two local densities. Their angular integrals at fixed r are

$$\int d\Omega \mathcal{H}_{\text{NY}}^V = -\frac{8\pi G^2}{r^4} \mathbf{P} \cdot \mathbf{S} + \mathcal{O}(r^{-5}). \quad (5.23)$$

These are angular integrals of the volume densities, not the codimension-two surface integrals; the latter contain the area factor r^2 . The pseudoscalar $\mathbf{P} \cdot \mathbf{S}$ is invariant under a spatial translation. Indeed,

$$x^i \longrightarrow x^i - a^i, \quad \mathbf{S} \longrightarrow \mathbf{S} - \mathbf{a} \times \mathbf{P}, \quad \mathbf{P} \cdot \mathbf{S} \longrightarrow \mathbf{P} \cdot \mathbf{S}. \quad (5.24)$$

It has the kinematic form of a spin projection along the momentum, but for a massive system it is the temporal component of the Pauli-Lubanski pseudovector. Thus this term is a frame-dependent boosted contribution, rather than an "extra helicity". It vanishes in the center-of-mass frame. We will discuss this result using boosted Kerr solution.

2. In the center-of-mass frame, $P_i = 0$, the leading results are

$$\mathcal{H}_{\text{NY}}^V = -\frac{6G^2}{r^5} S_i D_{\langle ij \rangle} n_j + \mathcal{O}(r^{-6}), \quad (5.25a)$$

$$\mathcal{H}_{\text{CS}}^V = \frac{24G^2}{r^7} \left[S_i D_{\langle ij \rangle} n_j + 2(S_i n_i) D_{\langle jk \rangle} n_j n_k \right] + \mathcal{O}(r^{-8}). \quad (5.25b)$$

Both angular integrals vanish because the displayed angular structures are odd under $\mathbf{n} \rightarrow -\mathbf{n}$. The leading contribution is a coupling between the current-dipole moment S_i and the time derivative of the mass quadrupole $I_{\langle ij \rangle}$.

The nonzero exterior value of $\mathcal{H}_{\text{CS}}^V$ in (5.25b) is not a contradiction. Due to the non-local projector (5.4), the projected constraint

$$\nabla^2 \Xi_i = -16\pi G T_{0i}^T \quad (5.26)$$

does not imply $\nabla^2 \Xi_i = 0$ outside the support of the unprojected stress tensor. The apparent tension arises only if the nonlocal projected source is treated as though it had the same compact support as $T_{\mu\nu}$.

Spin-2 modes. We now turn to the radiative tensor sector. In the slow motion approximation and at leading order in the wave zone expansion, the transverse-traceless field is

$$h_{ij}^{\text{TT}}(t, \mathbf{x}) = \frac{2G}{r} \Pi_{ij,kl}(\mathbf{n}) \ddot{I}_{kl}(u) + \dots, \quad u = t - r \quad (5.27)$$

in the slow motion limit. Here I_{ij} is the STF quadrupole moment of the source. Equation (5.27) is simultaneously the leading $1/r$ term and the leading mass quadrupole term in the slow motion multipole expansion. The ellipsis therefore includes both subleading powers of $1/r$ and higher radiative multipoles, the latter of which can also occur at order $1/r$.

To discuss the causal support on a constant time hypersurface, first consider an idealized source that is stationary before u_i and after u_f . Stationarity does not require the quadrupole itself to vanish. Rather, one may write

$$I_{ij}(u) = \begin{cases} I_{ij}^{\text{in}}, & u < u_i, \\ I_{ij}^{\text{dyn}}(u), & u_i < u < u_f, \\ I_{ij}^{\text{out}}, & u > u_f, \end{cases} \quad I_{ij}^{(p)} = 0 \quad \text{outside } (u_i, u_f), \quad p \geq 1. \quad (5.28)$$

This compact support assumption concerns the radiative time derivatives. It is stronger than necessary for the helicity densities below, but makes the causal discussion transparent. We also assume sufficient smoothness at u_i and u_f , so that the idealized switching does not introduce distributional wave fronts.

Consider the constant time hypersurface

$$V = \{t = \text{constant}\}, \quad t > u_f. \quad (5.29)$$

The outgoing null hypersurfaces $u = u_i$ and $u = u_f$ intersect V on spheres of radii

$$r_i = t - u_i, \quad r_f = t - u_f, \quad r_f < r_i. \quad (5.30)$$

The part of V on which the wave zone approximation is valid is consequently divided into

$r < r_f$	$\iff$	$u > u_f,$	post-radiation region,
$r_f < r < r_i$	$\iff$	$u_i < u < u_f,$	active-radiation-shell,
$r > r_i$	$\iff$	$u < u_i,$	pre-radiation region.

(5.31)

Every line in (5.31) is to be intersected with the wave zone condition

$$r \gg r_c, \quad r \gg \lambda_{\text{GW}}, \quad r \gg GM. \quad (5.32)$$

In particular, the first region need not contain a wave zone portion unless r_f is sufficiently large.

At fixed t , smaller radii correspond to later retarded times, whereas larger radii probe earlier states of the source. Thus

$$r \longrightarrow \infty \quad \text{at fixed finite } t \quad \implies \quad u = t - r \longrightarrow -\infty. \quad (5.33)$$

Spatial infinity on such a slice therefore lies in the pre-radiation region. In Figure 1, we have shown the three regions. For a compact pulse in linearized gravity on four-dimensional Minkowski spacetime, this sharp shell follows from Huygens propagation. On black hole backgrounds, curvature backscattering produces late time tails, as in Price's analysis of perturbations of Schwarzschild spacetime [52]. The sharp compact shell used here should therefore be regarded as a Minkowski space idealization rather than as a generic description of propagation on curved backgrounds.

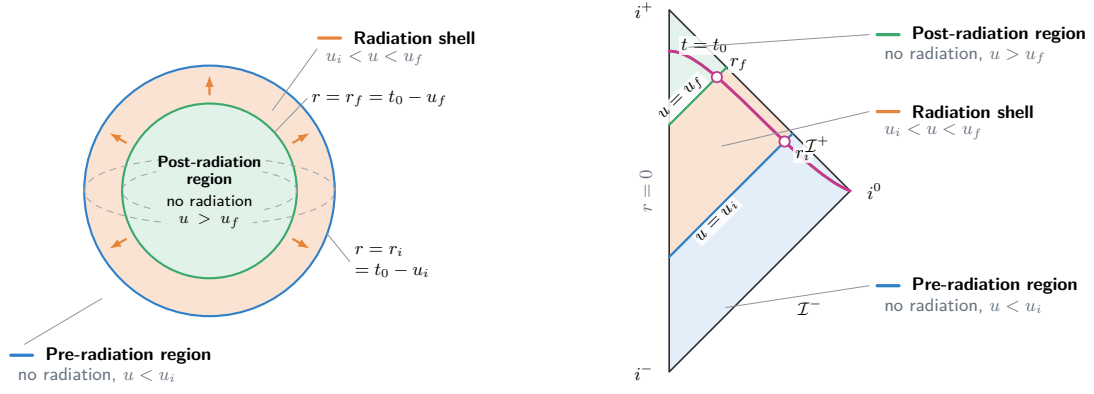


Figure 1: Intersection of a constant time hypersurface with the outgoing null hypersurfaces $u = u_i$ and $u = u_f$. The displayed three region decomposition applies to the finite duration radiative approximation.

Since we focus on linearized gravity, only in the radiation-shell, the helicity densities are non-vanishing. In this case, we have

$$\mathcal{H}_{NY}^T = \frac{G^2}{r^2} \ddot{I}_{ij} \ddot{I}_{kl} Q_{ijkl}(\mathbf{n}) + \mathcal{O}(r^{-3}), \quad (5.34a)$$

$$\mathcal{H}_{CS}^T = \frac{4G^2}{r^2} \ddot{I}_{ij} \ddot{I}_{kl} Q_{ijkl}(\mathbf{n}) + \mathcal{O}(r^{-3}). \quad (5.34b)$$

The tensor structure $Q_{ijkl}(\mathbf{n})$ has been defined in [53]

$$Q^{ijkl}(\mathbf{n}) = -P^{jk}(\mathbf{n}) \epsilon^{ilm} n_m. \quad (5.35)$$

The expression (5.34a) matches the helicity flux density near null infinity [53] up to some factors

$$\frac{r^2}{8\pi G} \mathcal{H}_{NY}^T = \frac{dH}{du d\Omega} + \mathcal{O}(r^{-1}). \quad (5.36)$$

The factor r^2 originates from the spatial volume element d^3x and converts the volume density into a radial density per unit solid angle

$$d^3x = r^2 dr d\Omega. \quad (5.37)$$

On a constant time hypersurface one has

$$du = -dr. \quad (5.38)$$

The minus sign is accounted for by reversing the radial integration limits. The factor $1/(8\pi G)$ is the canonical gravitational normalization appearing in the helicity flux density at $\mathcal{I}^+$.

The relation between the spin-2 gravitomagnetic helicity density and the helicity flux density should be understood as an asymptotic matching relation in their common domain of validity, rather than as an exact identity at arbitrary finite t and r . The quadrupole expression for $\mathcal{H}_{NY}^T$ is valid in the radiation zone, whereas $dH/(du d\Omega)$ is defined in the asymptotic region near $\mathcal{I}^+$. Their common matching region is therefore

$$\mathcal{D}_{\text{match}} = \{(t, r, \Omega) \mid r \gg r_c, \quad r \gg \lambda_{\text{GW}}, \quad r \gg GM, \quad u_i < u = t - r < u_f\}, \quad (5.39)$$

where r_c is the characteristic source size, λ_{GW} is the gravitational wavelength, and $u_i < u < u_f$ is the retarded-time interval during which the source radiates.

Within this overlap region, both descriptions are determined by the same leading radiative coefficient (5.27) and one obtains the asymptotic matching relation (5.36). More precisely, the equality of the leading asymptotic coefficients is defined by the null limit

$$\frac{dH}{du d\Omega}(u, \Omega) = \lim_{\substack{r \rightarrow \infty, t \rightarrow \infty \\ u=t-r \text{ finite}}} \frac{r^2}{8\pi G} \mathcal{H}_{\text{NY}}^T(t, r, \Omega), \quad u_i < u < u_f. \quad (5.40)$$

This limit should not be confused with the spatial infinity limit $r \rightarrow \infty$ at fixed finite t . In the latter case, $u = t - r \rightarrow -\infty$, and the observation point leaves the radiation-shell and enters the pre-radiation region. The complete multipole expansion [49] contains higher radiative multipoles without altering this correspondence.

The spin-2 gravito-current density has the same quadrupolar structure as the spin-2 gravitomagnetic density, with I_{ij} replaced by $\dot{I}_{ij}$. The latter can be compared with the gravitational helicity flux density at $\mathcal{I}^+$ considered in [53]. With the same rescaling, the spin-2 gravito-current density gives

$$\lim_{\substack{r, t \rightarrow \infty \\ u=t-r \text{ finite}}} \frac{r^2}{8\pi G} \mathcal{H}_{\text{CS}}^T = \frac{G}{2\pi} \ddot{I}_{ij} \ddot{I}_{kl} Q_{ijkl}(\mathbf{n}). \quad (5.41)$$

This is not the same helicity flux density studied in [53]. Instead, it agrees, up to normalization, with the density of the quantity studied in [54]. It is a higher time derivative, helicity sensitive quantity and differs in physical dimension from $dH/(dud\Omega)$ by two inverse powers of time. The comparison is valid in the common wave zone region and becomes an exact asymptotic matching only in the null limit $r, t \rightarrow \infty$ with $u = t - r$ held fixed.

For a monochromatic outgoing waveform of angular frequency ω , the leading wave-zone expressions obey

$$\mathcal{H}_{\text{CS}}^T = 4\omega^2 \mathcal{H}_{\text{NY}}^T. \quad (5.42)$$

For a circular binary, the leading gravitational wave frequency is $\omega_{\text{GW}} = 2\Omega$, and hence

$$\mathcal{H}_{\text{CS}}^T = 16\Omega^2 \mathcal{H}_{\text{NY}}^T. \quad (5.43)$$

For a general transient or multifrequency waveform there is no universal pointwise proportionality. The factor $4\omega^2$ applies frequency by frequency and, in the absence of coherent cross-frequency terms, to the corresponding time averaged spectral components.

For a finite duration signal, define

$$\overline{\mathcal{H}_{\text{NY}}^T}(r, \mathbf{n}) \equiv \frac{1}{u_f - u_i} \int_{u_i}^{u_f} du \mathcal{H}_{\text{NY}}^T(u, r, \mathbf{n}), \quad (5.44a)$$

$$\overline{\mathcal{H}_{\text{CS}}^T}(r, \mathbf{n}) \equiv \frac{1}{u_f - u_i} \int_{u_i}^{u_f} du \mathcal{H}_{\text{CS}}^T(u, r, \mathbf{n}). \quad (5.44b)$$

For a periodic source, $u_f - u_i$ is replaced by one period. Whenever the denominator is nonzero and the ratio is positive, one may define

$$\omega_\star^2(r, \mathbf{n}) \equiv \frac{\overline{\mathcal{H}_{\text{CS}}^T}(r, \mathbf{n})}{4 \overline{\mathcal{H}_{\text{NY}}^T}(r, \mathbf{n})}, \quad \lambda_\star(r, \mathbf{n}) \equiv \frac{2\pi}{\omega_\star(r, \mathbf{n})}. \quad (5.45)$$

For a monochromatic waveform, $\omega_\star = |\omega|$ and $\lambda_\star = 2\pi/|\omega|$. Both helicity densities are signed. Opposite chiralities at different frequencies can therefore make the denominator vanish or make the ratio negative. In that situation $\omega_\star$ is not a real characteristic frequency; one must instead define the frequency moment separately in each helicity sector, or adopt an explicitly positive spectral weighting.

6 Applications

In this section, we evaluate the four gauge invariant helicity densities introduced above for several representative gravitational configurations. The examples are chosen to probe complementary physical regimes: sourced and source-free fields, nonradiative and radiative sectors, and configurations carrying orbital, intrinsic-spin, or topological structure. Together, they clarify which parts of the gravitational field are measured by the spin-1 and spin-2 gravitomagnetic helicities and by their higher derivative gravito-current counterparts. We first consider two-body systems in the weak field and slow motion regime. This example connects the helicity densities with the source multipole moments and separates the constrained spin-1 field from the radiative transverse traceless field. In particular, it allows us to study the angular distribution of the spin-2 densities in the wave zone and their dependence on the orbital parameters. We then turn to a boosted Kerr black hole, which provides a nonradiative test of how linear momentum and intrinsic angular momentum enter the helicity densities in a chosen asymptotic inertial frame. The third example is a gyratonic pp-wave, namely a null gravitational field sourced by a beam that carries both energy and intrinsic angular momentum. It illustrates how spin-1 and spin-2 SVT sectors can coexist in a propagating configuration. Finally, we examine the gravitational Hopfion, a localized source free gravitational wave with nontrivial linked tendex and vortex structures. In this case the asymptotically decaying vacuum constraints eliminate the spin-1 SVT mode, so that only the two spin-2 helicity densities remain.

Throughout this section, all helicity densities are defined on constant time hypersurfaces and are evaluated only after the complete SVT projection has been performed. In particular, a coordinate component h_{0i} , or a gravitoelectromagnetic vector potential introduced in a specific coordinate system, must not be identified directly with the gauge invariant vector mode Ξ_i . For exact solutions, such as Kerr and pp-wave geometries, the present definitions apply to their linearization about Minkowski spacetime in the region where the weak field approximation is valid.

6.1 Two-body systems

We now evaluate the four local helicity densities for Newtonian two-body motion². The calculation is performed at leading order in the weak field and slow motion expansions. We neglect intrinsic spins, radiation reaction, periastron precession, and all post-Newtonian corrections, so the relative orbit is an exact Keplerian conic. For a bound binary this should be understood as the leading adiabatic approximation over one orbital period [55, 56].

The two stars, with masses M_1 and M_2 , move in the x - y plane and are separated by a distance D . In the center-of-mass frame, we use e to denote the orbital eccentricity: $e = 0$ for circular orbits, $0 < e < 1$ for elliptical orbits, $e = 1$ for parabolic orbits, and $e > 1$ for hyperbolic orbits. The total momentum of the system is zero and the orbital angular momentum points along $\mathbf{e}_z$. The observation point is in the direction

$$\mathbf{n} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta). \quad (6.1)$$

To simplify notation, we introduce

$$f \equiv 1 + e \cos \psi, \quad \mathbf{e}_r = (\cos \psi, \sin \psi, 0), \quad \mathbf{e}_\psi = (-\sin \psi, \cos \psi, 0). \quad (6.2)$$

Here ψ is the true anomaly, measured in the orbital plane from the direction of periastron. Our convention agrees with the standard Keplerian parametrization of [57]. The unit vectors $\mathbf{e}_r$ and $\mathbf{e}_\psi$ are the instantaneous radial and tangential unit vectors in the orbital plane. The instantaneous

²For an N -body system, closed-form trajectories are generally unavailable. Once the trajectories are specified, the corresponding local helicity densities follow from the formulas collected in Appendix C.

separation D between the two bodies is

$$D = \frac{\epsilon}{f} \quad (6.3)$$

where ϵ is called semi-latus rectum. It equals a for a circle, $a(1 - e^2)$ for an ellipse, remains finite in the parabolic limit, and equals $a(e^2 - 1)$ for a hyperbola. Here a is the orbital radius for a circular orbit, semi-major axis for an ellipse, positive magnitude of the conventional semi-major axis for a hyperbola.

The moment conventions require some care. Earlier we defined P_i , D_{ij} , and S_i using T_{0i} . With signature $(-+++)$, $T_{0i} = -T^{0i}$ at Newtonian order. Consequently, for those definitions,

$$P_i = -\sum_A M_A v_{Ai} = 0, \quad (6.4a)$$

$$S_i = -\sum_A M_A (\mathbf{x}_A \times \mathbf{v}_A)_i = -\mu \sqrt{G\bar{M}} \epsilon \delta_{iz}, \quad (6.4b)$$

$$I_{ij} = \mu D^2 \left(e_{ri} e_{rj} - \frac{1}{3} \delta_{ij} \right), \quad (6.4c)$$

$$D_{\langle ij \rangle} = -\frac{1}{2} \dot{I}_{ij}. \quad (6.4d)$$

The vanishing of P_i follows from the center-of-mass choice. The quantity S_i is the current dipole and equals the conserved minus orbital angular momentum. I_{ij} is the STF mass quadrupole moment. We have also defined the total mass of the binary

$$\bar{M} = M_1 + M_2 \quad (6.5)$$

and the reduced mass governing the relative Newtonian motion

$$\mu = \frac{M_1 M_2}{M_1 + M_2}. \quad (6.6)$$

The direction cosines adapted to the instantaneous orbit are

$$n_r \equiv \mathbf{n} \cdot \mathbf{e}_r = \sin \theta \cos(\phi - \psi), \quad (6.7a)$$

$$n_\psi \equiv \mathbf{n} \cdot \mathbf{e}_\psi = \sin \theta \sin(\phi - \psi), \quad (6.7b)$$

$$n_z \equiv \mathbf{n} \cdot \mathbf{e}_z = \cos \theta, \quad (6.7c)$$

and they satisfy $n_r^2 + n_\psi^2 + n_z^2 = 1$.

Two spin-1 helicity densities are

$$\mathcal{H}_{NY}^V = \frac{2G^3 \bar{M} \mu^2 \epsilon}{r^5} \frac{e \sin \psi}{f} n_z + \mathcal{O}(r^{-6}), \quad (6.8a)$$

$$\mathcal{H}_{CS}^V = \frac{24G^3 \bar{M} \mu^2 \epsilon}{r^7} n_z \left[\frac{e \sin \psi}{f} (2n_r^2 - 1) + 2n_r n_\psi \right] + \mathcal{O}(r^{-8}). \quad (6.8b)$$

The radiative spin-2 densities are

$$\mathcal{H}_{NY}^T = \frac{4G^{9/2} \bar{M}^{5/2} \mu^2}{r^2 \epsilon^{7/2}} n_z f^3 \mathcal{F}_{NY}(e; \psi, \mathbf{n}) + \mathcal{O}(r^{-3}), \quad (6.9a)$$

$$\mathcal{H}_{CS}^T = \frac{16G^{11/2} \bar{M}^{7/2} \mu^2}{r^2 \epsilon^{13/2}} n_z f^6 \mathcal{F}_{CS}(e; \psi, \mathbf{n}) + \mathcal{O}(r^{-3}) \quad (6.9b)$$

where

$$\begin{aligned}\mathcal{F}_{NY}(e; \psi, \mathbf{n}) &= 2(1 + n_z^2) + e \left[(6 - 2n_r^2 - 4n_\psi^2) \cos \psi - n_r n_\psi \sin \psi \right] \\ &\quad + e^2 \left[(3 - n_r^2 - 2n_\psi^2) \cos^2 \psi + n_r^2 - 1 - n_r n_\psi \sin \psi \cos \psi \right],\end{aligned}\quad (6.10a)$$

$$\begin{aligned}\mathcal{F}_{CS}(e; \psi, \mathbf{n}) &= 8(1 + n_z^2) + e \left[(30 - 14n_r^2 - 16n_\psi^2) \cos \psi - 4n_r n_\psi \sin \psi \right] \\ &\quad + e^2 \left[(15 - 7n_r^2 - 8n_\psi^2) \cos^2 \psi + n_r^2 - 1 - 4n_r n_\psi \sin \psi \cos \psi \right].\end{aligned}\quad (6.10b)$$

Equations (6.8a)–(6.9b) are the four master densities. The orbit classes differ only in the allowed eccentricity, the semi-latus rectum, and the range of the true anomaly. One should keep in mind that the vector and tensor results are not taken in the same asymptotic limit. This distinction is essential. The spin-1 expressions are large radius expansions on a fixed t hypersurface. They require $r \gg D(t)$, where D is the binary separation, and the orbital phase is evaluated at t . The spin-2 expressions are wave zone expansions with r, t large and $u = t - r$ fixed. Their orbital phase and source multipoles are evaluated at retarded time u . To avoid cumbersome notation, the same symbol ψ is used in both sectors. It means $\psi(t)$ in every formula carrying the superscript V and $\psi(u)$ in every formula carrying the superscript T .

Circular orbit. For a circular orbit,

$$e = 0, \quad \epsilon = a, \quad D = a, \quad \psi = \Omega\tau + \psi_0, \quad \Omega = \sqrt{\frac{GM}{a^3}}. \quad (6.11)$$

Here $\tau = t$ in the vector sector and $\tau = u$ in the tensor sector. The four densities reduce to

$$\mathcal{H}_{NY}^V = \mathcal{O}(r^{-6}), \quad (6.12a)$$

$$\mathcal{H}_{CS}^V = \frac{48G^3 \overline{M} \mu^2 a}{r^7} n_r n_\psi n_z + \mathcal{O}(r^{-8}), \quad (6.12b)$$

$$\mathcal{H}_{NY}^T = \frac{8G^{9/2} \overline{M}^{5/2} \mu^2}{r^2 a^{7/2}} n_z (1 + n_z^2) + \mathcal{O}(r^{-3}), \quad (6.12c)$$

$$\mathcal{H}_{CS}^T = \frac{128G^{11/2} \overline{M}^{7/2} \mu^2}{r^2 a^{13/2}} n_z (1 + n_z^2) + \mathcal{O}(r^{-3}). \quad (6.12d)$$

The spin-1 gravitomagnetic helicity density vanishes at the leading order. The spin-1 gravito-current helicity density may equivalently be written as

$$\mathcal{H}_{CS}^V = \frac{24G^3 \overline{M} \mu^2 a}{r^7} \sin^2 \theta \cos \theta \sin[2(\phi - \psi)] + \mathcal{O}(r^{-8}). \quad (6.13)$$

The helicity densities for the spin-2 modes are axisymmetric. They obey the useful monochromatic relation (5.43). Note that the spin-2 gravitomagnetic helicity density matches gravitational helicity flux density of [53] via the identity (5.40). At the leading order, the spin-2 helicity densities are independent of time while the spin-1 gravito-current helicity density oscillates with time. In a period, the average spin-1 gravito-current helicity density vanishes

$$\overline{\mathcal{H}_{CS}^V} = \mathcal{O}(r^{-8}). \quad (6.14)$$

Elliptic orbit. For an elliptic orbit,

$$0 < e < 1, \quad \epsilon = a(1 - e^2), \quad 0 \leq \psi < 2\pi, \quad \Omega = \sqrt{\frac{GM}{a^3}}. \quad (6.15)$$

Unlike the circular orbit, the instantaneous expressions retain the full phase dependence. One can use the period average to smooth out the time dependence. The period average must include the nonuniform Kepler measure $d\tau = d\psi/\dot{\psi}$

$$\overline{H} = \frac{1}{T} \int_0^T d\tau \mathcal{H} = \frac{1}{T} \int_0^{2\pi} \frac{d\psi}{\dot{\psi}} \mathcal{H}. \quad (6.16)$$

Here T is the period of the orbit. Therefore, the averaged helicity densities are

$$\overline{\mathcal{H}_{NY}^V} = \mathcal{O}(r^{-6}), \quad (6.17a)$$

$$\overline{\mathcal{H}_{CS}^V} = \mathcal{O}(r^{-8}), \quad (6.17b)$$

$$\overline{\mathcal{H}_{NY}^T} = \frac{G^{9/2} \overline{M}^{5/2} \mu^2}{2r^2 a^{7/2} (1-e^2)^2} n_z \times \left[16(1+n_z^2) + e^2 \{ 14(1+n_z^2) + 5\Delta \} \right] + \mathcal{O}(r^{-3}), \quad (6.17c)$$

$$\begin{aligned} \overline{\mathcal{H}_{CS}^T} = & \frac{G^{11/2} \overline{M}^{7/2} \mu^2}{8r^2 a^{13/2} (1-e^2)^5} n_z \times \left[(1+n_z^2) (1024 + 7328e^2 + 5232e^4 + 276e^6) \right. \\ & \left. + \Delta (464e^2 + 640e^4 + 51e^6) \right] + \mathcal{O}(r^{-3}). \end{aligned} \quad (6.17d)$$

Here $\Delta \equiv n_x^2 - n_y^2 = \sin^2 \theta \cos 2\phi$. The averaged spin-2 gravitomagnetic helicity density agrees, up to the normalization in Eq. (5.36), with the averaged helicity flux density of [53]. To our knowledge, the angular distribution of the averaged spin-2 gravito-current density for a binary has not previously been given. Both averaged densities above contain a common factor n_z . Consequently, their defining ratio (5.45) is a 0/0 expression on the orbital equator $n_z = 0$. For $n_z \neq 0$, cancellation of the common factor gives

$$\omega_*^2(\mathbf{n}) = \frac{\Omega^2}{16(1-e^2)^3} \frac{(1+n_z^2)(1024 + 7328e^2 + 5232e^4 + 276e^6) + \Delta(464e^2 + 640e^4 + 51e^6)}{16(1+n_z^2) + e^2[14(1+n_z^2) + 5\Delta]}. \quad (6.18)$$

The right-hand side of Eq. (6.18) is smooth at $n_z = 0$ and will be used below as the continuous angular extension of the off-equator ratio. When $e \rightarrow 0$, the characteristic frequency becomes $\omega_*(\mathbf{n}) = 2\Omega$. For small e , the characteristic frequency can be expanded as

$$\omega_*^2(\mathbf{n}) = 4\Omega^2 \left[1 + c_2(\mathbf{n})e^2 + \mathcal{O}(e^4) \right]. \quad (6.19)$$

The coefficient $c_2(\mathbf{n})$ is

$$c_2(\mathbf{n}) = \frac{297}{32} + \frac{9}{64} \frac{\Delta}{1+n_z^2}. \quad (6.20)$$

Since

$$|\Delta| = |n_x^2 - n_y^2| \leq n_x^2 + n_y^2 = 1 - n_z^2, \quad \Rightarrow \quad -1 \leq \frac{\Delta}{1+n_z^2} \leq 1, \quad (6.21)$$

we find $c_2(\mathbf{n})$ is always positive. More precisely,

$$\frac{585}{64} \leq c_2(\mathbf{n}) \leq \frac{603}{64}. \quad (6.22)$$

Therefore, for small e , the characteristic frequency

$$\omega_*(\mathbf{n}) > 2\Omega. \quad (6.23)$$

We have numerically checked that the right hand side of (6.18) is always positive for $0 < e < 1$. The positivity of (6.18) can also be established analytically. Set

$$x = e^2, \quad z = n_z^2, \quad u = \frac{\Delta}{1+z} = \frac{1-z}{1+z} \cos 2\phi, \quad |u| \leq 1. \quad (6.24)$$

Then

$$W(x, u) \equiv \frac{\omega_\star^2}{4\Omega^2} = \frac{1}{64(1-x)^3} \frac{A_E(x) + B_E(x)u}{C_E(x) + 5xu}, \quad (6.25)$$

where

$$A_E(x) = 1024 + 7328x + 5232x^2 + 276x^3, \quad (6.26a)$$

$$B_E(x) = 464x + 640x^2 + 51x^3, \quad (6.26b)$$

$$C_E(x) = 16 + 14x. \quad (6.26c)$$

For $0 < x < 1$,

$$C_E + 5xu \geq 16 + 9x > 0, \quad A_E + B_E u \geq A_E - B_E > 0,$$

the continuous extension satisfies $W > 0$. The dependence of W on u is linear fractional, and

$$\frac{\partial W}{\partial u} = \frac{x(2304 - 19904x - 16384x^2 - 666x^3)}{64(1-x)^3[C_E(x) + 5xu]^2}. \quad (6.27)$$

Therefore $\partial W/\partial u = 0$ does not locate a generic interior extremum. Instead, its sign determines which endpoint of the allowed interval in u gives the maximum. The nontrivial zero occurs at

$$x_c \simeq 0.106397, \quad e_c \simeq \sqrt{x_c} = 0.32619. \quad (6.28)$$

Since the global values $u = \pm 1$ are attained only at $z = 0$, the extrema of the continuous extension lie on the equator:

	$W_{\max}$	$W_{\min}$
$0 < e < e_c$	$(\theta, \phi) = (\frac{\pi}{2}, 0), (\frac{\pi}{2}, \pi)$	$(\theta, \phi) = (\frac{\pi}{2}, \frac{\pi}{2}), (\frac{\pi}{2}, \frac{3\pi}{2})$
$e_c < e < 1$	$(\theta, \phi) = (\frac{\pi}{2}, \frac{\pi}{2}), (\frac{\pi}{2}, \frac{3\pi}{2})$	$(\theta, \phi) = (\frac{\pi}{2}, 0), (\frac{\pi}{2}, \pi)$

At $e = e_c$, W is independent of u , and hence of direction:

$$W(e_c) = \frac{1024 + 7328e_c^2 + 5232e_c^4 + 276e_c^6}{64(1 - e_c^2)^3(16 + 14e_c^2)} \simeq 2.3328, \quad \omega_\star^2(e_c) \simeq 9.3312\Omega^2, \quad \omega_\star(e_c) \simeq 3.0547\Omega. \quad (6.29)$$

Figure 2 displays the continuous angular extension of W for three representative eccentricities. We use normalized Mollweide coordinates.

$$X = \frac{\phi}{\pi} \cos v, \quad Y = \sin v, \quad 2v + \sin 2v = \pi \cos \theta, \quad (6.30)$$

with ϕ increasing from left to right. The center of the map is $(\theta, \phi) = (\pi/2, 0)$, the lateral boundaries are the common meridian $\phi = \pm\pi$, and $X = \pm 1/2$ on the equator corresponds to $\phi = \pm\pi/2$. The reversal of the color pattern between $e = 0.2$ and $e = 0.5, 0.8$ follows from the sign change in (6.27).

Hyperbolic orbit. For a hyperbolic orbit, the orbital parameters satisfy

$$e > 1, \quad \epsilon = a(e^2 - 1), \quad -A < \psi < A, \quad A = \arccos\left(-\frac{1}{e}\right), \quad (6.31)$$

where $a > 0$ denotes the magnitude of the hyperbolic semi-major axis. In terms of the incoming speed v_∞ and impact parameter b , these quantities are given by

$$a = \frac{GM}{v_\infty^2}, \quad e = \sqrt{1 + \frac{b^2 v_\infty^4}{G^2 M^2}}. \quad (6.32)$$

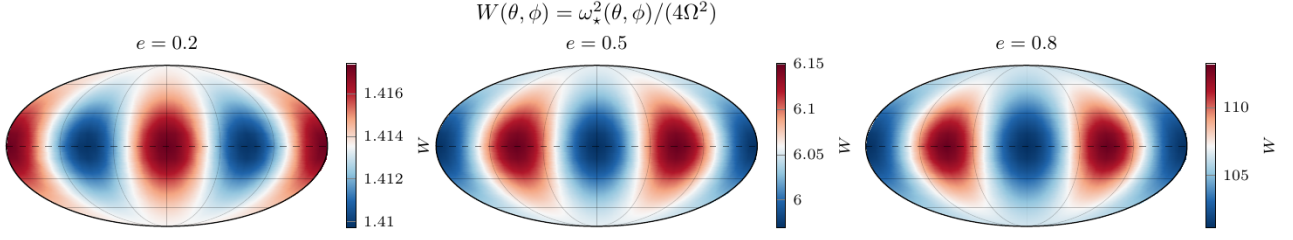


Figure 2: Continuous angular extension of $W = \omega_*^2/(4\Omega^2)$ for elliptic orbits. From left to right, $e = 0.2, 0.5, 0.8$. The horizontal dashed line is the orbital equator, and the remaining white curves are the Mollweide graticules. Each panel uses an independent color scale to display the angular pattern; colors should therefore not be compared as common absolute magnitudes across panels.

These are the standard Newtonian scattering relations and applications of the quadrupole formalism to unbound encounters may be found, for example, in [58].

The four helicity densities are time-dependent, and no finite orbital period exists in the hyperbolic case. The accumulated helicity over the entire scattering process may be defined as

$$\mathfrak{A}(r, \mathbf{n}) \equiv \int_{-\infty}^{+\infty} d\tau \mathcal{H}(\tau, r, \mathbf{n}). \quad (6.33)$$

With this definition, one obtains

$$\mathfrak{A}_{\text{NY}}^T(r, \mathbf{n}) = \frac{G^4 \bar{M}^2 \mu^2}{2r^2 \epsilon^2} n_z \kappa_{\text{NY}} + \mathcal{O}(r^{-3}), \quad (6.34)$$

$$\mathfrak{A}_{\text{CS}}^T(r, \mathbf{n}) = \frac{G^5 \bar{M}^3 \mu^2}{60r^2 \epsilon^5 e^2} n_z [A\mathcal{P}_A + \chi\mathcal{P}_\chi] + \mathcal{O}(r^{-3}), \quad (6.35)$$

where

$$\begin{aligned} \kappa_{\text{NY}} = \frac{2}{3e^2} & \left[A \{ 15e^4 \Delta + 6e^2(7e^2 + 8)(1 + n_z^2) \} \right. \\ & \left. + \chi \{ (12e^4 + e^2 + 2)\Delta + 6e^2(2e^2 + 13)(1 + n_z^2) \} \right], \end{aligned} \quad (6.36)$$

and the polynomials $\mathcal{P}_A$ and $\mathcal{P}_\chi$ are defined as

$$\mathcal{P}_A = 15e^4(51e^4 + 640e^2 + 464)\Delta + 60e^2(69e^6 + 1308e^4 + 1832e^2 + 256)(1 + n_z^2), \quad (6.37)$$

$$\mathcal{P}_\chi = (5891e^6 + 10882e^4 + 568e^2 - 16)\Delta + 20e^2(1969e^4 + 6562e^2 + 1864)(1 + n_z^2). \quad (6.38)$$

The parameter χ is defined as $\chi = \sqrt{e^2 - 1}$. For the hyperbolic orbit, we may also calculate characteristic frequency $\omega_{*,\text{hyp}}^2(\mathbf{n})$ and show that it is always well defined for any $e > 1$. Define

$$W_H(\theta, \phi) \equiv \frac{\omega_{*,\text{hyp}}^2(\theta, \phi)}{4\Omega_H^2}, \quad \Omega_H^2 = \frac{GM}{a^3},$$

then the maxima of W_H is located at $(\theta, \phi) = (\frac{\pi}{2}, \pm\frac{\pi}{2})$ for each fixed $e > 1$. The minima of W_H is located at $(\theta, \phi) = (\frac{\pi}{2}, 0)$ and $(\frac{\pi}{2}, \pi)$. We denote the maxima of W_H as $W_{H,\text{max}}(e)$ and the minima of W_H as $W_{H,\text{min}}(e)$. They are monotonically decreasing functions of e , as shown in Figure 3. $W_{H,\text{max}}(e)$ and $W_{H,\text{min}}(e)$ obey the following power law when $e \rightarrow \infty$

$$W_{H,\text{max}}(e) = \frac{25}{64} \frac{1}{e^2} + \mathcal{O}(e^{-3}), \quad (6.39a)$$

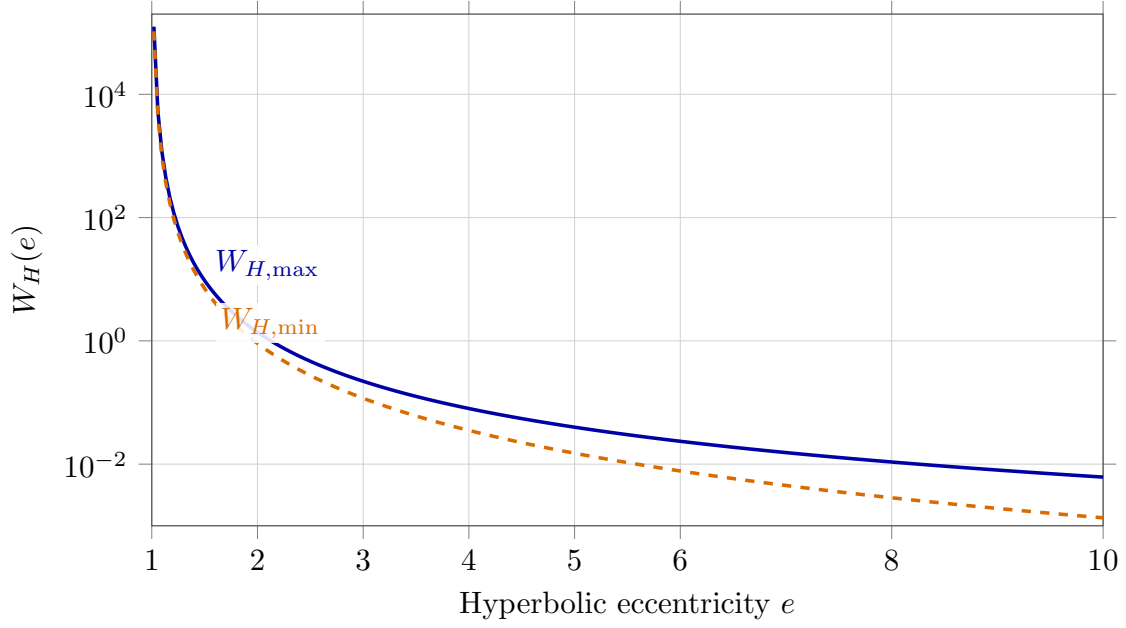


Figure 3: Extrema of the characteristic frequency ratio W_H for hyperbolic orbits.

$$W_{H,\min}(e) = \frac{327\pi}{1024} \frac{1}{e^3} + \mathcal{O}(e^{-4}). \quad (6.39b)$$

This interprets the discrepancy of the maxima and minima as $e \rightarrow \infty$. The large e limits obtained above require some explanation. In terms of the eccentricity e and the pericenter distance r_p , the pericenter speed is

$$v_p^2 = \frac{G\bar{M}(1+e)}{r_p}.$$

The weak field and slow motion conditions are

$$\frac{G\bar{M}}{r_p} \ll 1, \quad v_p^2 = \frac{G\bar{M}(1+e)}{r_p} \ll 1.$$

For a hyperbolic orbit, $e > 1$, the second condition is stronger and implies the first one. At fixed r_p , the Newtonian result therefore applies only in the finite interval

$$1 < e \ll \frac{r_p}{G\bar{M}} - 1. \quad (6.40)$$

This interval is nonempty and can be parametrically large when $r_p \gg G\bar{M}$, so that large but finite eccentricities can consistently be considered. However, the strict limit $e \rightarrow \infty$ at fixed r_p eventually gives $v_p \sim 1$ and leaves the domain of validity. A controlled $e \rightarrow \infty$ limit is possible only as a correlated scaling limit in which r_p also increases while

$$\frac{G\bar{M}(1+e)}{r_p} \ll 1$$

remains fixed and small. For example, one may set

$$r_p = \frac{G\bar{M}}{\nu^2}(1+e), \quad \nu \ll 1, \quad (6.41)$$

which gives $v_p^2 = \nu^2$ and $G\bar{M}/r_p \rightarrow 0$ as $e \rightarrow \infty$. Thus, the formal large e asymptotic expressions should be understood either within the finite Newtonian window at fixed r_p , or in this correlated weak field scaling limit.

We do not integrate the spin-1 large r formulas over the complete unbound orbit. At fixed observation radius, $D(t) \rightarrow \infty$ as $|t| \rightarrow \infty$, so the condition $r \gg D(t)$ eventually fails. Any divergence obtained by integrating spin helicity densities term by term is therefore an artifact of using a nonuniform spatial multipole expansion outside its domain. An exact treatment would require the unexpanded spin-1 field or a matched asymptotic construction.

Parabolic orbit. The parabolic orbit is obtained by taking the limits $e \rightarrow 1$, $A \rightarrow \pi$, and $\chi \rightarrow 0$ in the hyperbolic results. In this limit, one finds

$$\mathfrak{A}_{\text{NY}}^T = \frac{5\pi G^4 \overline{M}^2 \mu^2}{r^2 \epsilon_{\text{P}}^2} n_z \left[\Delta + 6(1 + n_z^2) \right], \quad (6.42)$$

$$\mathfrak{A}_{\text{CS}}^T = \frac{1155\pi G^5 \overline{M}^3 \mu^2}{4r^2 \epsilon_{\text{P}}^5} n_z \left[\Delta + 12(1 + n_z^2) \right]. \quad (6.43)$$

Here ϵ_{P} denotes the semi-latus rectum of the parabolic orbit. It should be noted that, for a parabolic orbit, one has $a \rightarrow \infty$ and $e \rightarrow 1$; to obtain a finite parabolic limit, the semi-latus rectum $\epsilon_{\text{P}} = a(e^2 - 1)$ must be kept finite. This quantity is given by

$$\epsilon_{\text{P}} = \frac{L^2}{GM\mu^2}, \quad (6.44)$$

where L is the orbital angular momentum of the parabolic orbit.

6.2 Boosted Kerr

In this subsection, the word boosted Kerr means a finite Lorentz transformation is applied to the Kerr geometry. This is not an accelerating Kerr black hole and it does not generate gravitational radiation. It is the same stationary spacetime described relative to a different asymptotic inertial frame and a different family of equal-time hypersurfaces. In the literature, there are two different treatment of the boosted Kerr solution: the construction of Burinskii and Magli [59], and the Lorentz-boosted Cartesian Kerr-Schild solution used by Huq, Choptuik, and Matzner [60]. The former gives an analytic description for an arbitrary magnitude and orientation of the boost, with particular emphasis on the principal null congruence, the singular set, and ultrarelativistic limits. The latter gives explicit real Cartesian metric and 3+1 data suited to numerical apparent-horizon calculations. For every finite boost the two constructions describe the same Kerr spacetime in different variables and with different normalizations of the Kerr-Schild null one-form. We will use the latter to derive the gauge invariant vector mode Ξ_i in the weak field, large distance, linear spin regime. The result is controlled by the ADM momentum and the intrinsic Kerr angular momentum.

A Kerr-Schild metric is written as [61]

$$g_{\mu\nu} = \eta_{\mu\nu} + 2H\ell_\mu\ell_\nu, \quad \eta^{\mu\nu}\ell_\mu\ell_\nu = 0. \quad (6.45)$$

The decomposition is invariant under

$$\ell_\mu \longrightarrow w\ell_\mu, \quad H \longrightarrow w^{-2}H. \quad (6.46)$$

For a Kerr spin along the positive z axis,

$$H = \frac{GM r^3}{r^4 + a^2 z^2}, \quad (6.47)$$

$$\ell_\mu = \left(1, \frac{rx + ay}{r^2 + a^2}, \frac{ry - ax}{r^2 + a^2}, \frac{z}{r}\right), \quad (6.48)$$

where r is defined implicitly by

$$\frac{x^2 + y^2}{r^2 + a^2} + \frac{z^2}{r^2} = 1. \quad (6.49)$$

Here M is the rest mass and $J = Ma$ is the rest-frame Kerr angular momentum. If barred coordinates are comoving with the hole, the boosted fields are obtained from

$$H(x) = \bar{H}(\Lambda^{-1}x), \quad \ell_\mu(x) = \frac{\partial \bar{x}^{\bar{\alpha}}}{\partial x^\mu} \bar{\ell}_{\bar{\alpha}}(\Lambda^{-1}x). \quad (6.50)$$

The main result of [60] is the direct $3 + 1$ decomposition

$$ds^2 = -\alpha^2 dt^2 + \gamma_{ij}(dx^i + \beta^i dt)(dx^j + \beta^j dt) \quad (6.51)$$

with

$$\alpha = \left(1 + 2H\ell_t^2\right)^{-1/2}, \quad (6.52)$$

$$\beta_i = 2H\ell_t\ell_i, \quad (6.53)$$

$$\gamma_{ij} = \delta_{ij} + 2H\ell_i\ell_j. \quad (6.54)$$

These fields were used to test an apparent-horizon finder for unboosted and boosted Schwarzschild and Kerr holes. The horizon is coordinate distorted and Lorentz contracted, while its geometrical area remains invariant. In the rest frame, the weak field limit gives

$$H = \frac{GM}{r} + \mathcal{O}(a^2/r^3), \quad (6.55)$$

$$\ell_0 = 1, \quad (6.56)$$

$$\ell_i = n_i - \frac{(\mathbf{a} \times \mathbf{n})_i}{r} + \mathcal{O}(a^2/r^2). \quad (6.57)$$

The absence of a term linear in a in H follows from Eq. (6.47). Choose the laboratory frame so that the hole moves with velocity $\mathbf{v}$. To first order in v ,

$$\bar{t} = t - \mathbf{v} \cdot \mathbf{x}, \quad \bar{\mathbf{x}} = \mathbf{x} - \mathbf{v}t. \quad (6.58)$$

With

$$\mathbf{R} = \mathbf{x} - \mathbf{v}t, \quad r = |\mathbf{R}|, \quad \mathbf{n} = \mathbf{R}/r, \quad (6.59)$$

the boosted null one-form is

$$\ell_0 = 1 - \mathbf{v} \cdot \mathbf{n} + \mathcal{O}(v^2, av), \quad (6.60)$$

$$\ell_i = n_i - v_i - \frac{(\mathbf{a} \times \mathbf{n})_i}{r} + \mathcal{O}(v^2, a^2, av). \quad (6.61)$$

Substitution into $h_{\mu\nu} = 2H\ell_\mu\ell_\nu$ gives

$$h_{0i} = \frac{2GM}{r}n_i - \frac{2GM}{r}[v_i + n_i(\mathbf{v} \cdot \mathbf{n})] - \frac{2G}{r^2}(\mathbf{J} \times \mathbf{n})_i + \dots. \quad (6.62)$$

The first term is longitudinal because

$$\frac{n_i}{r} = \partial_i \ln r. \quad (6.63)$$

The other two displayed fields are transverse for $r \neq 0$

$$\partial_i \left\{ \frac{v_i + n_i(\mathbf{v} \cdot \mathbf{n})}{r} \right\} = 0, \quad \partial_i \left\{ \frac{(\mathbf{J} \times \mathbf{n})_i}{r^2} \right\} = 0. \quad (6.64)$$

Thus the transverse spin-1 mode is

$$\Xi_i = -\frac{2G}{r} \left[P_i^{\text{ADM}} + n_i(\mathbf{P}^{\text{ADM}} \cdot \mathbf{n}) \right] - \frac{2G}{r^2} \epsilon_{ijk} J_j n_k + \mathcal{O}(r^{-3}, v^2 J, a^2), \quad (6.65)$$

where $\mathbf{P}^{\text{ADM}} = M\mathbf{v} + \mathcal{O}(v^3)$ at this stage of the expansion. Comparing with (5.20), we should identify

$$\mathbf{P} = -\mathbf{P}^{\text{ADM}}, \quad \mathbf{S} = -\mathbf{J} \quad (6.66)$$

and $D_{ij} = 0$. The minus sign comes from the choice of convention. Thus, the spin-1 gravitomagnetic helicity density and the spin-1 gravito-current helicity density are

$$\mathcal{H}_{\text{NY}}^V = -\frac{G^2}{r^4} \left[\mathbf{P} \cdot \mathbf{S} + 3(\mathbf{P} \cdot \mathbf{n})(\mathbf{S} \cdot \mathbf{n}) \right] + \mathcal{O}(r^{-5}), \quad (6.67a)$$

$$\mathcal{H}_{\text{CS}}^V = \frac{4G^2}{r^6} \left[\mathbf{P} \cdot \mathbf{S} + 3(\mathbf{P} \cdot \mathbf{n})(\mathbf{S} \cdot \mathbf{n}) \right] + \mathcal{O}(r^{-7}). \quad (6.67b)$$

These are local, frame-dependent equal-time densities. Their nonzero value in a boosted frame does not signal gravitational radiation. A global Lorentz transformation of Kerr has vanishing Bondi news, so the radiative $1/r$ tensor sector vanishes

$$h_{ij}^{\text{TT}} = 0, \quad \mathcal{H}_{\text{NY}}^T = \mathcal{H}_{\text{CS}}^T = 0 \quad (6.68)$$

in the wave zone. Although the exact boosted Kerr geometry exists for any finite boost, the formulas above are valid under the conditions

$$GM/r \ll 1, \quad |a|/r \ll 1, \quad v \ll 1. \quad (6.69)$$

These formulas retain the terms at first order in the weak field and the leading parts that are separately linear in v and a . The displayed vJ helicity is their quadratic interference within this truncation. No global helicity integral through the black hole interior is evaluated. An integral on an exterior slice would require the corresponding inner boundary term.

6.3 Gyratonic pp-wave

Gyratonic pp-waves are Brinkmann geometries [62] generated by null matter carrying intrinsic angular momentum [63–65]. They provide idealized gravitational fields of spinning radiation beam pulses and ultrarelativistic spinning sources. In this subsection we will review the exact geometry and then retain its first-order weak field part, which is the order required for the quadratic helicity densities studied in this work.

Let $A, B = 1, 2$ label the flat transverse coordinates and define

$$u = \frac{t - z}{\sqrt{2}}, \quad v = \frac{t + z}{\sqrt{2}}. \quad (6.70)$$

The Brinkmann form of a four-dimensional gyratonic pp-wave is

$$ds^2 = -2dudv + \delta_{AB} dx^A dx^B + 2m_A(u, \mathbf{x}_\perp) dudx^A + H(u, \mathbf{x}_\perp) du^2. \quad (6.71)$$

The vector ∂_v is a null Killing vector. The metric (6.71) is invariant under the Brinkmann gauge transformation

$$v \longrightarrow v + \xi(u, \mathbf{x}_\perp), \quad m_A \longrightarrow m_A - \partial_A \xi, \quad H \longrightarrow H - 2\partial_u \xi. \quad (6.72)$$

Thus m_A alone is not a gauge invariant observable. Its transverse field strength

$$f_{AB} = \partial_A m_B - \partial_B m_A \quad (6.73)$$

is gauge invariant. On a non-contractible exterior region, the circulation $\oint m_A dx^A$ can also carry gauge invariant global information [65, 66].

For a gyratonic source whose only nonzero components are

$$T_{uu} = T_{uu}(u, \mathbf{x}_\perp), \quad T_{uA} = T_{uA}(u, \mathbf{x}_\perp), \quad (6.74)$$

the relevant exact Ricci components are

$$R_{uA} = \frac{1}{2} \partial^B f_{AB}, \quad (6.75)$$

$$R_{uu} = -\frac{1}{2} \Delta_\perp H + \partial_u \partial^A m_A + \frac{1}{4} f_{AB} f^{AB}, \quad (6.76)$$

where $\Delta_\perp = \partial^A \partial_A$. In the transverse Brinkmann gauge $\partial^A m_A = 0$, the first order Einstein equations reduce to

$$\Delta_\perp H = -16\pi G T_{uu}, \quad \Delta_\perp m_A = -16\pi G T_{uA}. \quad (6.77)$$

The term $f_{AB} f^{AB}$ contributes to the exact metric at second order in the source angular momentum. It can be omitted consistently here because the helicity densities are already quadratic in the first order metric. A second order correction to H would first modify them at cubic order. Gyratonic pp-waves belong to the VSI (vanishing scalar invariants) spacetime. Namely, all scalar polynomial curvature invariants vanish [67, 68]. A smooth matter-filled gyratonic region is generically of Petrov type III, whereas a smooth exterior vacuum region is of type N [65, 66]. This does not imply that the foliation dependent, equal time quadratic helicity densities defined below must vanish. In particular, these densities are not four-dimensional scalar polynomial curvature invariants.

In polar coordinates $x = \rho \cos \phi$ and $y = \rho \sin \phi$,

$$ds^2 = d\rho^2 + \rho^2 d\phi^2 - 2dudv + 2m_\phi(u, \rho, \phi) dud\phi + H(u, \rho, \phi) du^2. \quad (6.78)$$

We model the matter as a smooth gyratonic null fluid whose stress tensor reproduces, at leading paraxial and geometric optics order, the cycle averaged energy and spin angular momentum fluxes of a finite width, finite duration, circularly polarized Gaussian electromagnetic beam pulse [64]. This is a controlled phenomenological model of the leading Maxwell stress tensor, rather than a globally exact, non-diffracting Maxwell solution. Let w denote the effective transverse intensity radius and set

$$\chi \equiv \frac{\rho^2}{w^2}. \quad (6.79)$$

We choose

$$T_{uu} = \frac{\sqrt{2} \mathcal{M}(u)}{\pi w^2} e^{-\chi}, \quad T_{uA} = \frac{\mathcal{J}(u)}{\pi w^4} \epsilon_{AB} x_B e^{-\chi}, \quad \epsilon_{12} = +1. \quad (6.80)$$

This source is conserved at the order under consideration because $\partial^A T_{uA} = 0$.

For a circularly polarized beam, let $\lambda_\gamma = \pm 1$ denote the photon helicity. We choose the energy and spin current envelopes to satisfy

$$\mathcal{J}(u) = \lambda_\gamma \frac{\mathcal{M}(u)}{\omega_0}. \quad (6.81)$$

This is consistent with the integrated relation $J = \lambda_\gamma E/\omega_0$ derived for a circularly polarized radiation beam pulse of [64]. A smooth pulse envelope is

$$\mathcal{M}(u) = \frac{E}{\sqrt{\pi} \tau} e^{-u^2/\tau^2}, \quad \mathcal{J}(u) = \lambda_\gamma \frac{E}{\sqrt{\pi} \tau \omega_0} e^{-u^2/\tau^2}, \quad (6.82)$$

so that $\int du \mathcal{M} = E$ is the energy of the beam. The geometric optics and paraxial conditions are

$$\omega_0 w \gg 1, \quad \omega_0 \tau \gg 1. \quad (6.83)$$

The rigid transverse profile is meaningful only near the beam waist and over propagation distances small compared with the Rayleigh scale $z_R = \mathcal{O}(\omega_0 w^2)$.

Define

$$F(\chi) = \gamma_E + \ln \chi - \text{Ei}(-\chi), \quad A(\chi) = 1 - e^{-\chi}. \quad (6.84)$$

Here $\text{Ei}(x)$ is the exponential integral

$$\text{Ei}(x) = \text{PV} \int_{-\infty}^x \frac{e^t}{t} dt \quad (6.85)$$

and γ_E is the Euler constant. The first order solution of Eq. (6.77) that is regular on the beam axis is

$$H(u, \rho) = -4\sqrt{2}G\mathcal{M}(u)F(\chi), \quad a = -4G\mathcal{J}(u)A(\chi)d\phi. \quad (6.86)$$

Thus

$$ds^2 = -2dudv + d\rho^2 + \rho^2 d\phi^2 - 8G\mathcal{J}(u)A(\chi)dud\phi - 4\sqrt{2}G\mathcal{M}(u)F(\chi)du^2. \quad (6.87)$$

The solution is smooth at $\rho = 0$. At large transverse radius,

$$H \sim -8\sqrt{2}G\mathcal{M}(u) \ln(\rho/w), \quad a \longrightarrow -4G\mathcal{J}(u)d\phi, \quad (6.88)$$

and the angular momentum profile is encoded in the circulation,

$$\mathcal{J}(u) = -\frac{1}{8\pi G} \oint a. \quad (6.89)$$

The logarithmic behavior in Eq. (6.88) shows that this idealized pp-wave is not asymptotically flat. Consequently, the inverse Laplacians entering the SVT projectors require an infrared boundary condition or a specified global completion of the beam. In the absence of such data, the formulas below are local, slowly varying representatives of the projected fields rather than a unique global SVT decomposition.

On a constant time hypersurface, the nonzero first-order metric components are

$$h_{00} = \frac{H}{2}, \quad h_{0A} = \frac{a_A}{\sqrt{2}}, \quad h_{0z} = -\frac{H}{2}, \quad h_{Az} = -\frac{a_A}{\sqrt{2}}, \quad h_{zz} = \frac{H}{2}. \quad (6.90)$$

The gauge invariant modes are obtained via the formula (2.11).

Closed local coordinate expressions are especially simple in the slowly varying regime

$$\frac{w}{\tau} \ll 1. \quad (6.91)$$

The leading transverse contributions to the gauge invariant vector and tensor SVT variables are

$$\begin{aligned}\Xi_\rho &= \mathcal{O}(w/\tau), \\ \Xi_\phi &= -2\sqrt{2}G\mathcal{J}(u)A(\chi) + \mathcal{O}(w^2/\tau^2), \\ \Xi_z &= 2\sqrt{2}G\mathcal{M}(u)F(\chi) + \mathcal{O}(w^2/\tau^2),\end{aligned}\tag{6.92}$$

and

$$\begin{aligned}h_{\rho\rho}^{\text{TT}} &= \frac{G\mathcal{M}(u)}{\sqrt{2}} \left[F(\chi) - 1 + \frac{A(\chi)}{\chi} \right] + \mathcal{O}(w^2/\tau^2), \\ h_{\phi\phi}^{\text{TT}} &= \frac{G\mathcal{M}(u)\rho^2}{\sqrt{2}} \left[F(\chi) + 1 - \frac{A(\chi)}{\chi} \right] + \mathcal{O}(w^2/\tau^2), \\ h_{zz}^{\text{TT}} &= -\sqrt{2}G\mathcal{M}(u)F(\chi) + \mathcal{O}(w^2/\tau^2), \\ h_{\phi z}^{\text{TT}} &= h_{z\phi}^{\text{TT}} = 2\sqrt{2}G\mathcal{J}(u)A(\chi) + \mathcal{O}(w^2/\tau^2), \\ h_{\rho z}^{\text{TT}} &= \mathcal{O}(w/\tau), \\ h_{\rho\phi}^{\text{TT}} &= h_{\phi\rho}^{\text{TT}} = \mathcal{O}(w/\tau).\end{aligned}\tag{6.93}$$

The omitted terms contain longitudinal derivatives and are suppressed by powers of w/τ . The local expressions displayed below therefore represent the leading slowly varying approximation to the gauge invariant SVT variables, rather than the exact global SVT projection of the ideal pp-wave metric.

With the spin-1 and spin-2 modes, one obtains

$$\mathcal{H}_{\text{NY}}^V = \frac{4\lambda_\gamma G^2 E^2}{\pi\omega_0\tau^2} e^{-2u^2/\tau^2} \left[\frac{F(\chi)e^{-\chi}}{w^2} - \frac{A(\chi)^2}{\rho^2} \right] + \mathcal{O}(w^2/\tau^2),\tag{6.94a}$$

$$\mathcal{H}_{\text{CS}}^V = \frac{32\lambda_\gamma G^2 E^2}{\pi\omega_0\tau^2 w^4} e^{-2u^2/\tau^2 - \chi} + \mathcal{O}(w^2/\tau^2),\tag{6.94b}$$

$$\mathcal{H}_{\text{NY}}^T = \frac{\lambda_\gamma G^2 E^2}{2\pi\omega_0\tau^2 w^2} e^{-2u^2/\tau^2} \mathcal{G}_T(\chi) + \mathcal{O}(w^2/\tau^2),\tag{6.94c}$$

$$\mathcal{H}_{\text{CS}}^T = \frac{4\lambda_\gamma G^2 E^2 u^2}{\pi\omega_0\tau^6 w^2} e^{-2u^2/\tau^2} \mathcal{G}_T(\chi) + \mathcal{O}(w^4/\tau^4),\tag{6.94d}$$

where

$$\mathcal{G}_T(\chi) = [-3F(\chi) - 1 + q(\chi)][2e^{-\chi} - q(\chi)] + q(\chi)[8A(\chi) - 3F(\chi) + 1 - q(\chi)]\tag{6.95}$$

with

$$q(\chi) = \frac{A(\chi)}{\chi}.\tag{6.96}$$

Thus, in the slow envelope expansion $w/\tau \ll 1$, the leading contributions scale as

$$(\mathcal{H}_{\text{NY}}^V, \mathcal{H}_{\text{CS}}^V, \mathcal{H}_{\text{NY}}^T, \mathcal{H}_{\text{CS}}^T) = (O(1), O(1), O(1), O(w^2/\tau^2)).\tag{6.97}$$

All odd powers of w/τ vanish for the axisymmetric rigid-profile configuration considered here. Therefore, the first corrections to the first three densities are of order $O(w^2/\tau^2)$, whereas the first correction to $\mathcal{H}_{\text{CS}}^T$ is of order $O(w^4/\tau^4)$. All four densities reverse sign when the photon helicity λ_γ is reversed. Since a Gaussian has exponentially small but nonzero support at every finite ρ , the spin-1 gravito-current density is not exactly zero outside a sharp beam radius. Instead, it is exponentially localized in the beam core.

For the spin-2 density, a monochromatic TT mode obeys

$$\mathcal{H}_{\text{CS}}^T = 4\omega^2 \mathcal{H}_{\text{NY}}^T.\tag{6.98}$$

Thus the pointwise characteristic frequency depends on time

$$\omega_{*,p}^2 \equiv \frac{\mathcal{H}_{\text{CS}}^T}{4\mathcal{H}_{\text{NY}}^T} = \frac{2u^2}{\tau^4}. \quad (6.99)$$

Its retarded time average is

$$\omega_*^2 = \frac{\int_{-\infty}^{+\infty} du \mathcal{H}_{\text{CS}}^T}{4 \int_{-\infty}^{+\infty} du \mathcal{H}_{\text{NY}}^T} = \frac{1}{2\tau^2}. \quad (6.100)$$

Hence the accumulated spin-2 ratio measures the inverse duration of the gravitational waveform,

$$\omega_* = \frac{1}{\sqrt{2}\tau}, \quad (6.101)$$

rather than the optical carrier frequency ω_0 . The carrier frequency controls the amplitude through $\mathcal{J} = \lambda_\gamma \mathcal{M}/\omega_0$ and cancels from both characteristic ratios.

The derivative hierarchy between the two helicities in each spin sector can be used to define a characteristic scale. For a transverse vector helicity eigenmode satisfying $\nabla \times \Xi_\sigma = \sigma k \Xi_\sigma$, the definition of spin-1 helicity densities gives

$$\mathcal{H}_{\text{CS}}^V = -2k^2 \mathcal{H}_{\text{NY}}^V. \quad (6.102)$$

The corresponding local characteristic reduced wave length is therefore

$$\lambda_*^2 = -\frac{2\mathcal{H}_{\text{NY}}^V}{\mathcal{H}_{\text{CS}}^V}. \quad (6.103)$$

For the Gaussian gyraton profile this becomes

$$\lambda_*^2 = \frac{w^2 [A(\chi)^2/\chi - F(\chi)e^{-\chi}]}{4e^{-\chi}}. \quad (6.104)$$

This characteristic wave length probes the transverse structure and is of order w away from zeros of $\mathcal{H}_{\text{NY}}^V$.

Note that the two-dimensional transverse logarithmic Green function implies

$$\mathcal{H}_{\text{NY}}^V \sim -\frac{4G^2 \mathcal{M} \mathcal{J}}{\rho^2}, \quad \mathcal{H}_{\text{NY}}^T \sim \frac{5G^2 \mathcal{M} \mathcal{J}}{\rho^2}, \quad \rho \gg w. \quad (6.105)$$

Consequently, the corresponding transverse integrals are infrared sensitive and require an outer matching radius since the ideal four-dimensional pp-wave is not asymptotically flat in the transverse direction.

6.4 Gravitational Hopfion

The term gravitational Hopfion does not identify a unique metric field. Several related, but inequivalent, constructions occur in the literature. Penrose transform constructions generate massless spin-2 fields whose electric and magnetic Weyl tensors possess linked tendex and vortex structures. The type N solution and its relation to linked gravitational radiation were studied in [69, 70]. Moreover, type N, D, and III Hopfions were subsequently classified in [71]. In [72], it is shown that a reduced complex

scalar can be used to reconstruct the gauge invariant spin-2 data and generalized gravitational Hopfions. The complex anti-self-dual Kerr-Schild solutions obtained from Plebanski's second-heavenly construction [73] include the Sparling-Tod metric [74, 75]. Its single copy is the electromagnetic Hopfion, and its curvature realizes a Weyl double copy construction [76]. These gravitational Hopfions are direct generalizations of the Hopfions in electromagnetism [77] and various other branches of physics [78, 79].

We set the characteristic length of the Hopfion to unity. Introduce

$$D_+ = r^2 - (t + i)^2 = r^2 + d^2, \quad d = 1 - it, \quad \ell_i = (x + iy, -i(x + iy), z - id) \quad (6.106)$$

and the complex spatial seed ³

$$q_{ij}(t, \mathbf{x}) = \frac{4\kappa \ell_i \ell_j}{D_+^3}, \quad \kappa \in \mathbb{R}. \quad (6.107)$$

It carries the $p = q = 1$ principal spinor and denominator structure of the complex anti-self-dual Kerr-Schild Hopfion. The linearized regime further requires $|h_{ij}^{\text{TT}}| \ll 1$, which, under the unit length convention, implies taking the amplitude κ to be sufficiently small. The metric field is kept at $O(\kappa)$, and the helicity densities at $O(\kappa^2)$. We do not assert that $\eta_{\mu\nu} + h_{\mu\nu}$ is the corresponding exact nonlinear Einstein metric. The physical tensor field is generated by the seed q_{ij}

$$h_{ij}^{\text{TT}} = \text{Re} \left[\hat{\Pi}_{ij,kl} q_{kl} \right]. \quad (6.108)$$

The complex seed $q_{ij} = O(\kappa)$ is the spatial part of the perturbation obtained by linearizing the complex type N Kerr-Schild Hopfion solution about its flat background. It is not by itself a complete four-dimensional metric perturbation and is not yet transverse and traceless. The field used in the calculations is produced only after the spatial transverse and traceless projection and the real-part operation in Eq. (6.108). Use the Fourier convention

$$\tilde{f}(\mathbf{k}) = \int d^3\mathbf{x} e^{-i\mathbf{k}\cdot\mathbf{x}} f(\mathbf{x}), \quad f(\mathbf{x}) = \int \frac{d^3\mathbf{k}}{(2\pi)^3} e^{i\mathbf{k}\cdot\mathbf{x}} \tilde{f}(\mathbf{k}). \quad (6.109)$$

For $\text{Re } d > 0$,

$$\mathcal{F} \left[(r^2 + d^2)^{-3} \right] = \frac{\pi^2}{4d^3} (1 + dk) e^{-dk}, \quad k = |\mathbf{k}|. \quad (6.110)$$

Multiplication by x_i in Eq. (6.107) becomes differentiation with respect to k_i . After those derivatives and the algebraic Fourier projector are applied, all longitudinal and trace pieces cancel, giving

$$\tilde{h}_{ij}^{\text{TT}}(t, \mathbf{k}) = -\pi^2 \kappa k e^{-k} \cos(kt) w_i(\mathbf{n}) w_j(\mathbf{n}), \quad \mathbf{n} = \frac{\mathbf{k}}{k}, \quad (6.111)$$

where

$$\mathbf{w}(\mathbf{n}) = (in_y - n_x n_z, -in_x - n_y n_z, 1 - n_z^2). \quad (6.112)$$

The polarization vector satisfies

$$n_i w_i = 0, \quad w_i w_i = 0, \quad i(\mathbf{n} \times \mathbf{w})_i = w_i, \quad w_i^* w_i = 2(1 - n_z^2). \quad (6.113)$$

³More explicitly, Eq. (1) and the $p = q = 1$ specialization stated immediately before Eq. (5) in [76] provide the Robinson principal spinor and the D_+^{-3} profile, respectively. Equations (10)–(11) of that reference embed the corresponding null Maxwell spinor in a complex anti-self-dual Kerr-Schild metric. After the imaginary-time translation $t \rightarrow t + i$ described below its Eq. (15), the spatial part linear in the Kerr-Schild parameter gives Eq. (6.107), up to the overall normalization absorbed into κ .

Consequently, Eq. (6.111) is transverse and traceless. It also obeys the vacuum wave equation. A useful exact inverse transform is

$$h_{ij}^{\text{TT}} = \text{Re} \left[-\frac{\kappa}{2} \mathcal{L}_i \mathcal{L}_j u(d, r) \right], \quad (6.114)$$

where

$$u(d, r) = -\frac{d}{r} \arctan\left(\frac{r}{d}\right) - \frac{1}{2} \log(d^2 + r^2), \quad (6.115)$$

and

$$\mathcal{L}_x = \partial_x \partial_z - \partial_d \partial_y, \quad \mathcal{L}_y = \partial_y \partial_z + \partial_d \partial_x, \quad \mathcal{L}_z = -(\partial_x^2 + \partial_y^2). \quad (6.116)$$

The branch in Eq. (6.115) is the analytic continuation from $\text{Re } d > 0$. Formally,

$$\tilde{u}(d, k) = \frac{2\pi^2 e^{-dk}}{k^3}. \quad (6.117)$$

The potential u by itself has an infrared ambiguity, but $\mathcal{L}_i \mathcal{L}_j u$ is infrared finite. Equation (6.115) is the representative selected by the Fourier boundary condition in Eq. (6.109). In particular, replacing its first term by $d \arctan(d/r)/r$ without the compensating $-\pi d/(2r)$ term changes the TT field.

At fixed t , Eq. (6.114) is smooth at the origin and falls as $h_{ij}^{\text{TT}} = \mathcal{O}(r^{-4})$. The surface terms generated by the spatial integrations by parts used below therefore vanish at spatial infinity.

For real κ , we have

$$\dot{h}_{ij}^{\text{TT}}(0, \mathbf{x}) = 0, \quad \mathcal{H}_{CS}^T(0, \mathbf{x}) = 0. \quad (6.118)$$

The spin-2 gravitomagnetic helicity density remains nonzero. Let

$$c = \frac{z}{r} = \cos \theta, \quad \alpha(r) = \arctan r. \quad (6.119)$$

The exact result is

$$\mathcal{H}_{NY}^T(0, \mathbf{x}) = \frac{\kappa^2}{16r^{10}(1+r^2)^4} \mathcal{N}(r, c), \quad (6.120)$$

where

$$\begin{aligned} \mathcal{N}(r, c) = & -18(1+r^2)^4(45c^4 - 10c^2 + 13)\alpha^2 + 12r(1+r^2)\alpha \left[18c^4r^6 + 171c^4r^4 + 360c^4r^2 + 135c^4 \right. \\ & \left. + 12c^2r^6 + 42c^2r^4 - 80c^2r^2 - 30c^2 + 18r^6 + 75r^4 + 104r^2 + 39 \right] \\ & + 2r^2 \left[108c^4r^6 - 801c^4r^4 - 1350c^4r^2 - 405c^4 - 440c^2r^6 - 302c^2r^4 + 300c^2r^2 + 90c^2 \right. \\ & \left. - 148r^6 - 385r^4 - 390r^2 - 117 \right], \end{aligned} \quad (6.121)$$

with $\alpha = \alpha(r)$. The apparent singularity in Eq. (6.120) is removable. Indeed,

$$\mathcal{H}_{NY}^T(0, \mathbf{x}) = \frac{96}{25} \kappa^2 + \mathcal{O}(r^2), \quad r \longrightarrow 0, \quad (6.122a)$$

$$\mathcal{H}_{NY}^T(0, \mathbf{x}) = \frac{9\pi\kappa^2}{4r^9} (3c^4 + 2c^2 + 3) + \mathcal{O}(r^{-10}), \quad r \longrightarrow \infty. \quad (6.122b)$$

Direct integration gives

$$H_{NY}^T = \int d^3\mathbf{x} \mathcal{H}_{NY}^T(0, \mathbf{x}) = \frac{\pi^2 \kappa^2}{2}. \quad (6.123)$$

For general t , we find the non-vanishing spin-2 gravitomagnetic and gravito-current helicities

$$H_{NY}^T(t) = \frac{\pi^2 \kappa^2}{4} \left[1 + \text{Re}(1 - it)^{-6} \right], \quad (6.124a)$$

$$H_{CS}^T(t) = \frac{21\pi^2 \kappa^2}{2} \left[1 - \text{Re}(1 - it)^{-8} \right]. \quad (6.124b)$$

We now verify the energy-helicity inequalities directly for the Hopfion defined by Eqs. (6.111)–(6.113). The positive frequency amplitude belongs to a single circular polarization sector. Introducing the normalized tensor

$$e_{ij}^{(+)}(\mathbf{n}) = \frac{w_i(\mathbf{n})w_j(\mathbf{n})}{\sqrt{2}(1 - n_z^2)}, \quad e_{ij}^{(+)*}e_{ij}^{(+)} = 2,$$

the mode amplitudes satisfy

$$|h_+(\mathbf{k})|^2 = \frac{\pi^4 \kappa^2}{2} k^2 e^{-2k} (1 - n_z^2)^2, \quad h_-(\mathbf{k}) = 0.$$

The negative frequency complex conjugate required to make the metric real does not constitute an independent opposite-helicity mode.

Using

$$\int d\Omega_{\mathbf{k}} (1 - n_z^2)^2 = \frac{32\pi}{15},$$

the angle-integrated energy spectrum becomes

$$\mathcal{E}_{\text{Hopf}}(k) = \frac{k^4}{8\pi G (2\pi)^3} \int d\Omega_{\mathbf{k}} |h_+(\mathbf{k})|^2 = \frac{\pi \kappa^2}{60G} k^6 e^{-2k}.$$

Consequently,

$$E_{\text{Hopf}} = \int_0^\infty dk \mathcal{E}_{\text{Hopf}}(k) = \frac{\pi \kappa^2}{60G} \frac{6!}{2^7} = \frac{3\pi \kappa^2}{32G}.$$

Since

$$\overline{\text{Re}(1 - it)^{-n}} = 0, \quad n > 1,$$

their long-time averages of the helicities are

$$\overline{H_{NY}^T} = \frac{\pi^2 \kappa^2}{4}, \quad \overline{H_{CS}^T} = \frac{21\pi^2 \kappa^2}{2}.$$

The right-hand side of the Nieh-Yan inequality (4.34a) evaluates to

$$8\pi G \int_0^\infty \frac{dk}{k} \mathcal{E}_{\text{Hopf}}(k) = 8\pi G \frac{\pi \kappa^2}{60G} \frac{5!}{2^6} = \frac{\pi^2 \kappa^2}{4} = \left| \overline{H_{NY}^T} \right|. \quad (6.125)$$

Similarly, one can verify (4.34b). The time average is essential. For example,

$$H_{NY}^T(0) = \frac{\pi^2 \kappa^2}{2},$$

which is twice the corresponding spectral bound $\pi^2 \kappa^2 / 4$. The additional contribution is the positive-negative-frequency interference term and is removed by the average. The inequalities therefore constrain the averaged or dephased helicities, not the instantaneous quantities in Eqs. (6.124a) and (6.124b).

The vector modes of the above gravitational Hopfion vanish. However, the absence of the spin-1 densities does not imply that the gravitational Hopfion lacks nontrivial linked structure. In the

Hopfion literature, the linking refers to the tendex and vortex structures determined by the electric and magnetic parts of the Weyl tensor, rather than to an independent SVT vector mode. The nonvanishing parity odd spin-2 pairings found here probe the chirality of the same radiative TT sector from which those curvature tensors are constructed. Nevertheless, H_{NY}^T and H_{CS}^T should not, without an additional topological argument, be identified directly with a Hopf invariant or a linking number.

7 Conclusion and discussion

In this work, we have investigated the helicity structures induced on a spacelike hypersurface by two four-dimensional topological densities: the Pontryagin and the Nieh-Yan density. Our construction combines the descent of these topological terms with the SVT decomposition of linearized gravity. It therefore provides a systematic way of separating the helicity structures carried by the gauge invariant spin-1 and spin-2 sectors of the gravitational field.

At quadratic order in the metric perturbations, the descendants of the Pontryagin and Nieh-Yan terms can be decomposed into gauge invariant bulk contributions and codimension-two surface terms. After the surface improvements have been separated, four helicity structures emerge

$$\mathcal{H}_{\text{NY}}^V = -\frac{1}{4}\Xi_i\epsilon_{ijk}\partial_j\Xi_k, \quad \mathcal{H}_{\text{NY}}^T = \frac{1}{4}h_{ij}^{\text{TT}}(Ch^{\text{TT}})_{ij}, \quad (7.1a)$$

$$\mathcal{H}_{\text{CS}}^V = -\frac{1}{2}\epsilon_{ijk}(\partial_j\Xi_k)\nabla^2\Xi_i, \quad \mathcal{H}_{\text{CS}}^T = \dot{h}_{ij}^{\text{TT}}(C\dot{h}^{\text{TT}})_{ij}. \quad (7.1b)$$

We have referred to the one derivative Nieh-Yan expressions as the spin-1 and spin-2 gravitomagnetic helicities, and to the higher derivative Pontryagin expressions as the corresponding gravito-current helicities.

The spin-2 gravito-current helicity density must be distinguished from the complete spin-2 Pontryagin descendant,

$$\mathcal{H}_{\text{CS},0}^T = \frac{1}{2} \left[\dot{h}_{ij}^{\text{TT}}(C\dot{h}^{\text{TT}})_{ij} + h_{ij}^{\text{TT}}\nabla^2(Ch^{\text{TT}})_{ij} \right]. \quad (7.2)$$

The replacement $h_{ij}^{\text{TT}} \rightarrow \dot{h}_{ij}^{\text{TT}}$ relates the Nieh-Yan induced spin-2 gravitomagnetic helicity density to the spin-2 gravito-current helicity density, but does not transform it directly into the complete Pontryagin descendant. Relating these two expressions requires the field equations, integrations by parts, and suitable boundary conditions.

Although these quantities descend from four-dimensional topological densities, their local representatives on a spatial hypersurface should not themselves be regarded as topological invariants. They are gauge invariant pseudoscalar densities at the linearized level, but their integrals can depend on the hypersurface, the boundary conditions, and the treatment of codimension-two surface terms. In particular, they need not be quantized and need not be conserved for a general sourced or time-dependent configuration.

We have analyzed these helicities from two complementary perspectives. The first is the plane-wave or mode expansion. This analysis makes particularly transparent the close correspondence between magnetic helicity and the spin-2 gravitomagnetic helicity. The second perspective is the source multipole expansion. In linearized Einstein gravity, the gauge invariant vector mode is determined by the transverse momentum density, whereas the transverse traceless tensor mode is sourced by T_{ij}^{TT} and propagates according to a retarded wave equation. The multipole expansion of Ξ_i relates the spin-1 densities to the total momentum, current dipole, quadrupole, and the remaining irreducible moments of the momentum distribution. The wave zone expansion of h_{ij}^{TT} , on the other hand,

relates the spin-2 densities to time derivatives of the source mass quadrupole and to higher radiative multipoles. Care is required because the transverse projections T_{0i}^T and T_{ij}^{TT} are spatially nonlocal and need not have compact support, even when the original stress tensor does.

These general results were illustrated by several physically distinct configurations. For two-body systems, the spin-1 densities probe the constrained field generated by orbital momentum and angular momentum, while the spin-2 densities describe the helicity distribution of quadrupolar gravitational radiation. Bound and unbound trajectories provide different notions of averaging: periodic averaging for elliptic orbits and accumulated helicity along the complete encounter for parabolic and hyperbolic orbits. The corresponding angular distributions also provide a natural setting for the helicity weighted characteristic frequency. A particularly noteworthy feature of the elliptic orbit result is the existence of a critical eccentricity

$$e_c \simeq 0.32619. \quad (7.3)$$

At $e = e_c$, the angular dependences of the period-averaged spin-2 gravitomagnetic and gravitocurrent helicity densities become proportional. Their ratio is therefore independent of the observation direction, and the normalized characteristic frequency satisfies

$$W(\theta, \phi) \equiv \frac{\omega_\star^2(\theta, \phi)}{4\Omega^2} = 2.33279. \quad (7.4)$$

The all-sky distribution of W is consequently uniform at the critical eccentricity, so that there is no distinguished direction of maximum or minimum characteristic frequency. This does not imply that either helicity density, or the gravitational radiation itself, is isotropic. Instead, their angular anisotropies cancel in the ratio defining $\omega_\star$. As e crosses e_c , the sign of the remaining angular-anisotropy coefficient reverses, interchanging the directions in which W attains its maxima and minima.

A boosted Kerr black hole provides a complementary nonradiative example. It shows how the total momentum and intrinsic spin of a compact object combine in the spin-1 sector when the field is described relative to a chosen asymptotic inertial frame. The gyratonic pp-wave illustrates a null configuration carrying both energy and intrinsic angular momentum, for which spin-1 and spin-2 SVT components can coexist. Interestingly, the two helicity sectors consequently probe complementary scales of the same source. The spin-1 characteristic length measures the transverse beam width w , whereas the spin-2 characteristic frequency measures the inverse pulse duration τ^{-1} . The energy E , Newton's constant G , the polarization λ_γ , and the carrier frequency ω_0 determine the amplitudes of the helicity densities but cancel from these characteristic scale ratios. The helicity ratios therefore separate the geometry of the source from its overall strength: the vector sector resolves its transverse vortical structure, while the tensor sector resolves its temporal variation. Finally, the type N gravitational Hopfion provides a localized source-free radiative configuration. Under the vacuum constraints and asymptotically decaying boundary conditions, its spin-1 mode vanishes and only the spin-2 helicity densities remain. These metric helicities are related to, but should not be identified directly with, the linking of the tendex and vortex fields constructed from the Weyl curvature.

Several directions deserve further investigation.

1. The relation between equal-time helicity densities and radiative helicity fluxes at $\mathcal{I}^+$ should be developed more systematically. A spatial density on a $t = \text{const}$ slice and a flux density on $\mathcal{I}^+$ are different observables. Their relation requires a controlled matching limit in the region where the wave zone and null infinity expansions are simultaneously valid.
2. The codimension-two surface terms should be studied for finite regions and open boundaries. The electromagnetic analogy suggests that a gravitational counterpart of relative helicity may be necessary when the relevant fields have nonvanishing flux through the boundary.

3. The four densities can be extracted from analytical waveforms and numerical binary black hole simulations. Their angular distributions and time dependence may provide interesting observations on circular polarization, precession, parity violation, and helicity memory that are complementary to the usual energy and angular momentum fluxes.
4. The relation between the Pauli-Lubanski operator and the spin-2 helicity density suggests a representation-theoretic extension to free massless fields of arbitrary integer spin. Let $\varphi_{i_1 \dots i_s}^{\text{TT}}$ be a symmetric, transverse, and traceless rank- s field on a spatial hypersurface. Its generalized symmetric curl may be defined by

$$(C\varphi)_{i_1 \dots i_s} = \epsilon_{kl(i_1} \partial_k \varphi_{i_2 \dots i_s)l}. \quad (7.5)$$

With the normalized symmetrization used here, the temporal Pauli-Lubanski operator satisfies

$$W_{(s)}^0 = sC. \quad (7.6)$$

On circularly polarized plane waves,

$$C\varphi^{(\pm s)} = \pm k \varphi^{(\pm s)}, \quad W_{(s)}^0 \varphi^{(\pm s)} = \pm s k \varphi^{(\pm s)}, \quad (7.7)$$

which reproduces the physical helicities $\pm s$. This observation naturally leads to a hierarchy of parity odd quadratic functionals,

$$H_s^{(n)} = \int d^3\mathbf{x} \varphi_{i_1 \dots i_s}^{\text{TT}} (-\nabla^2)^n (C\varphi^{\text{TT}})_{i_1 \dots i_s}, \quad n = 0, 1, \dots \quad (7.8)$$

The $n = 0$ member generalizes the magnetic or gravitomagnetic helicity, whereas the higher- n members measure progressively higher signed moments of the helicity spectrum. The cases $s = 1$ and $s = 2$ studied in this work are the first two members of this possible higher spin hierarchy. The Pauli-Lubanski construction alone, however, fixes only the helicity operator and its spectrum. It does not determine the normalization, conservation law, boundary completion, or topological origin of $H_s^{(n)}$. These questions should first be investigated for free Fronsdal fields and then, where possible, in frame-like higher spin theories on Minkowski or anti-de Sitter backgrounds.

A related open problem is whether the higher spin quadratic helicities can be obtained from the descent of genuine higher spin characteristic forms. For an ordinary gauge connection, Chern-Weil theory distinguishes clearly between a bulk characteristic class, its boundary Chern-Simons or transgression form, and the functional induced on a chosen hypersurface. For metric-like higher spin fields, by contrast, the generalized curvature carries several antisymmetric index pairs and does not automatically define an ordinary differential form characteristic class.

It is important to distinguish four notions in such an extension: a bulk topological characteristic class, a boundary descendant, a gauge invariant helicity functional, and an integer-valued linking or Hopf invariant. The results of the present work establish a relation between the first three notions for spin 1 and spin 2 under specified boundary conditions, but they do not imply that the resulting helicity integrals are integer-valued topological invariants. Establishing a genuine linking interpretation, especially for gravitational Hopfions and their higher spin analogues, remains an interesting open problem.

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A Boundary terms at spatial infinity

Here we consider a weak field and slowly moving binary system in its center-of-mass frame,

$$P_i = 0, \quad \int d^3\mathbf{x} x_i T_{00} = 0. \quad (\text{A.1})$$

The limit considered in this subsection is

$$t = \text{finite}, \quad r \longrightarrow \infty. \quad (\text{A.2})$$

It approaches spatial infinity i^0 , rather than $\mathcal{I}^+$, because

$$u = t - r \longrightarrow -\infty. \quad (\text{A.3})$$

At leading Newtonian order, the two gauge invariant scalar modes behave as

$$\Phi(t, \mathbf{x}) = -\frac{G\overline{M}}{r} + \mathcal{O}(r^{-3}), \quad (\text{A.4})$$

$$\Theta(t, \mathbf{x}) = \frac{2G\overline{M}}{r} + \mathcal{O}(r^{-3}), \quad (\text{A.5})$$

where $\overline{M} = M_1 + M_2$ is the total mass of the binary system. The mass-dipole terms are absent in the center-of-mass frame. Consequently,

$$\Phi = \mathcal{O}(r^{-1}), \quad \Theta = \mathcal{O}(r^{-1}), \quad (\text{A.6})$$

while their leading time-dependent contributions arise from the mass quadrupole:

$$\dot{\Phi}, \dot{\Theta} = \mathcal{O}(r^{-3}), \quad \partial_i \Phi, \partial_i \Theta = \mathcal{O}(r^{-2}). \quad (\text{A.7})$$

In a general inertial frame, the spin-1 mode has the leading behavior⁴

$$\Xi_i = \frac{2G}{r} [P_i + n_i(\mathbf{P} \cdot \mathbf{n})] + \mathcal{O}(r^{-2}). \quad (\text{A.8})$$

Thus, $\Xi_i = \mathcal{O}(r^{-1})$ when the total momentum is nonzero. In the center-of-mass frame, the r^{-1} term vanishes and one obtains

$$\Xi_i = \mathcal{O}(r^{-2}), \quad (\text{A.9})$$

together with

$$\partial_j \Xi_i = \mathcal{O}(r^{-3}), \quad \partial_j \partial_k \Xi_i = \nabla^2 \Xi_i = \mathcal{O}(r^{-4}). \quad (\text{A.10})$$

The leading wave zone expression for the spin-2 mode is

$$h_{ij}^{\text{TT}}(t, r\mathbf{n}) = \frac{2G}{r} \Pi_{ij,kl}(\mathbf{n}) \ddot{I}_{kl}(u) + \mathcal{O}(r^{-2}), \quad u = t - r. \quad (\text{A.11})$$

Its behavior at spatial infinity depends on the boundary data at $u \rightarrow -\infty$. For an idealized Keplerian binary that has been periodic for all past times,

$$\ddot{I}_{ij}(u) = \mathcal{O}(1), \quad (\text{A.12})$$

and hence

$$h_{ij}^{\text{TT}} = \mathcal{O}(r^{-1}). \quad (\text{A.13})$$

⁴One can find this expression in the next section.

However, $r h_{ij}^{\text{TT}}$ then oscillates as $r \rightarrow \infty$, and in general does not possess a smooth limit at i^0 . For an isolated system that becomes stationary in the remote past, one may instead impose

$$I_{ij}^{(2)}(u) \longrightarrow 0, \quad u \longrightarrow -\infty. \quad (\text{A.14})$$

It follows that

$$h_{ij}^{\text{TT}} = o(r^{-1}). \quad (\text{A.15})$$

With a regular asymptotic expansion, this behavior is usually strengthened to $\mathcal{O}(r^{-2})$ or faster. For the idealized, eternally periodic binary in the center-of-mass frame, the six modes can therefore be summarized as

$$(\Phi, \Theta, \Xi_{+1}, \Xi_{-1}, h_{+2}, h_{-2}) = (\mathcal{O}(r^{-1}), \mathcal{O}(r^{-1}), \mathcal{O}(r^{-2}), \mathcal{O}(r^{-2}), \mathcal{O}(r^{-1}), \mathcal{O}(r^{-1})). \quad (\text{A.16})$$

For an isolated system with stationary remote-past data, the last two entries should instead be replaced by $o(r^{-1})$.

We first impose regular spatial-infinity boundary conditions. In particular, the coefficient of the leading r^{-1} tensor field is assumed to be stationary as $u \rightarrow -\infty$. The relevant derivative falloffs are

$$h_{ij}^{\text{TT}} = \mathcal{O}(r^{-1}), \quad \partial_k h_{ij}^{\text{TT}} = \mathcal{O}(r^{-2}), \quad \partial_k \partial_l h_{ij}^{\text{TT}} = \mathcal{O}(r^{-3}), \quad (\text{A.17a})$$

$$\dot{h}_{ij}^{\text{TT}} = \mathcal{O}(r^{-2}), \quad \partial_k \dot{h}_{ij}^{\text{TT}} = \mathcal{O}(r^{-3}), \quad (\text{A.17b})$$

$$\Xi_i = \mathcal{O}(r^{-2}), \quad \partial_j \Xi_i = \mathcal{O}(r^{-3}), \quad \partial_j \partial_k \Xi_i = \mathcal{O}(r^{-4}), \quad (\text{A.17c})$$

$$\dot{\Theta} = \mathcal{O}(r^{-3}), \quad \partial_i \dot{\Theta} = \mathcal{O}(r^{-4}). \quad (\text{A.17d})$$

Substitution into the five boundary densities gives

$$\mathcal{B}_{\Xi\Xi}^r = \mathcal{O}(r^{-6}), \quad \mathcal{B}_{hh}^r = \mathcal{O}(r^{-4}), \quad (\text{A.18a})$$

$$\mathcal{B}_{\dot{\Theta}\Xi}^r = \mathcal{O}(r^{-6}), \quad \mathcal{B}_{\dot{h}\Xi}^r = \mathcal{O}(r^{-5}), \quad \mathcal{B}_{\Theta h}^r = \mathcal{O}(r^{-4}). \quad (\text{A.18b})$$

Since the area element on a large sphere is

$$dS = r^2 d\Omega, \quad (\text{A.19})$$

the corresponding surface contributions behave as

$$\frac{\mathcal{B}_{\Xi\Xi}^r}{r^2 \mathcal{B}^r} \Big| \frac{\mathcal{B}_{hh}^r}{\mathcal{O}(r^{-4})} \frac{\mathcal{B}_{\dot{\Theta}\Xi}^r}{\mathcal{O}(r^{-4})} \frac{\mathcal{B}_{\dot{h}\Xi}^r}{\mathcal{O}(r^{-3})} \frac{\mathcal{B}_{\Theta h}^r}{\mathcal{O}(r^{-2})}. \quad (\text{A.20})$$

They consequently vanish at spatial infinity:

$$\lim_{r \rightarrow \infty} \int_{S_r} dS \mathcal{B}_\alpha^r = 0, \quad \alpha \in \{\Xi\Xi, hh, \dot{\Theta}\Xi, \dot{h}\Xi, \Theta h\}. \quad (\text{A.21})$$

This conclusion does not follow from $h_{ij}^{\text{TT}} = \mathcal{O}(r^{-1})$ alone. For an eternally periodic binary,

$$h_{ij}^{\text{TT}} = \frac{F_{ij}(u, \mathbf{n})}{r} + \mathcal{O}(r^{-2}), \quad u = t - r, \quad (\text{A.22})$$

and radial derivatives acting on u do not improve the radial falloff:

$$\partial h^{\text{TT}}, \quad \partial^2 h^{\text{TT}}, \quad \dot{h}^{\text{TT}}, \quad \partial \dot{h}^{\text{TT}} = \mathcal{O}(r^{-1}). \quad (\text{A.23})$$

A conservative power counting then gives

$$\mathcal{B}_{\Xi\Xi}^r = \mathcal{O}(r^{-6}), \quad \mathcal{B}_{hh}^r = \mathcal{O}(r^{-2}), \quad (\text{A.24})$$

$$\mathcal{B}_{\Theta\Xi}^r = \mathcal{O}(r^{-6}), \quad \mathcal{B}_{h\Xi}^r = \mathcal{O}(r^{-3}), \quad \mathcal{B}_{\Theta h}^r = \mathcal{O}(r^{-3}). \quad (\text{A.25})$$

All surface terms still vanish except for the tensor-tensor term, which may have a finite limit. Its leading local contribution is

$$\mathcal{B}_{hh}^r = -\frac{1}{2r^2} \epsilon^{ijk} n_j F_{i\ell}(u, \mathbf{n}) \partial_u^2 F_{k\ell}(u, \mathbf{n}) + \mathcal{O}(r^{-3}). \quad (\text{A.26})$$

Therefore, the vanishing of the complete codimension-two boundary term requires a stationary remote past condition, a no incoming radiation condition with suitable regularity, or an explicit condition setting the tensor flux through spatial infinity to zero.

B Pauli-Lubanski operator and the helicity of spin-1 field

This section discuss the relation between Pauli-Lubanski operator and the magnetic helicity in electromagnetism. Applying the temporal Pauli-Lubanski component to a four-vector and subsequently performing the Helmholtz decomposition⁵ selects the transverse vector potential

$$(W^0 \mathbf{a}^T)_i = (\nabla \times \mathbf{a}^T)_i = b_i. \quad (\text{B.5})$$

This gives a gauge invariant, but spatially nonlocal, representative

$$\tilde{h}_m = \mathbf{a}^T \cdot \mathbf{b} \quad (\text{B.6})$$

of the usual magnetic helicity density. Its integral agrees with $\int \mathbf{a} \cdot \mathbf{b} d^3x$ for closed, periodic, or sufficiently decaying fields. The transverse field is obtained using the nonlocal projector

$$a_i^T = \hat{P}_{ij} a_j. \quad (\text{B.7})$$

With the gauge invariant density (B.6), we may define the integrated quantity,

$$\tilde{H}_m = \int_V d^3x \mathbf{a}^T \cdot \mathbf{b} = \int_V d^3x a_i^T (W_{(1)}^0 a^T)_i \quad (\text{B.8})$$

The standard magnetic helicity

$$H_m = \int_V d^3x \mathbf{a} \cdot \mathbf{b} \quad (\text{B.9})$$

plays a central role in force-free magnetic fields and in the topological description of linked flux tubes. Substituting $\mathbf{a} = \mathbf{a}^T + \nabla\psi$ gives

$$H_m = \int_V d^3x \mathbf{a}^T \cdot \mathbf{b} + \int_V d^3x \nabla\psi \cdot \mathbf{b} = \tilde{H}_m + \int_{\partial V} d^2y \sqrt{q} r_i \psi b^i, \quad (\text{B.10})$$

⁵On a fixed spatial hypersurface, decompose the electromagnetic field

$$a_0 = \varphi, \quad a_i = a_i^T + \partial_i \psi, \quad \partial_i a_i^T = 0. \quad (\text{B.1})$$

Under the gauge transformation

$$a_\mu \longrightarrow a_\mu + \partial_\mu \alpha, \quad (\text{B.2})$$

the variables transform according to

$$\delta\varphi = \dot{\alpha}, \quad \delta\psi = \alpha, \quad \delta a_i^T = 0. \quad (\text{B.3})$$

The reduced gauge invariant variables may thus be chosen as

$$\Psi = \varphi - \dot{\psi}, \quad a_i^T. \quad (\text{B.4})$$

where $\nabla \cdot \mathbf{b} = 0$ has been used. Whenever one of the following standard conditions holds:

$$\begin{aligned} r_i b_i \Big|_{\partial V} = 0 \quad &\text{or periodic boundary conditions,} \\ &\text{or sufficiently rapid falloff at spatial infinity.} \end{aligned} \quad (\text{B.11})$$

Equation (B.8) is therefore not a new global magnetic helicity. It is the transverse-potential representative of the standard functional (B.9).

At the density level,

$$\mathbf{a} \cdot \mathbf{b} - \mathbf{a}^T \cdot \mathbf{b} = \nabla \cdot (\psi \mathbf{b}). \quad (\text{B.12})$$

Thus $\mathbf{a} \cdot \mathbf{b}$ is local but gauge dependent, whereas $\mathbf{a}^T \cdot \mathbf{b}$ is gauge invariant after fixing the Helmholtz boundary problem but nonlocal as a functional of the original potential. This use of transverse potentials is standard in reduced phase space and duality based descriptions of electromagnetism [2, 3, 18, 80, 81]. Since

$$\nabla \times \mathbf{b} = \nabla \times (\nabla \times \mathbf{a}^T) = -\nabla^2 \mathbf{a}^T, \quad (\text{B.13})$$

it follows that

$$\mathbf{a}^T = -\nabla^{-2}(\nabla \times \mathbf{b}). \quad (\text{B.14})$$

On $\mathbb{R}^3$, with fields decaying at infinity, this becomes

$$\mathbf{a}^T(\mathbf{x}) = \frac{1}{4\pi} \int d^3 \mathbf{y} \frac{\mathbf{b}(\mathbf{y}) \times (\mathbf{x} - \mathbf{y})}{|\mathbf{x} - \mathbf{y}|^3}. \quad (\text{B.15})$$

The transverse representation of magnetic helicity can hence be written entirely in terms of $\mathbf{b}$ as the nonlocal Gauss-type integral

$$\widetilde{H}_m = \frac{1}{4\pi} \int d^3 \mathbf{x} d^3 \mathbf{y} \frac{\mathbf{b}(\mathbf{x}) \cdot [\mathbf{b}(\mathbf{y}) \times (\mathbf{x} - \mathbf{y})]}{|\mathbf{x} - \mathbf{y}|^3} = \frac{1}{4\pi} \int d^3 \mathbf{x} d^3 \mathbf{y} \frac{(\mathbf{b}(\mathbf{x}) \times \mathbf{b}(\mathbf{y})) \cdot (\mathbf{x} - \mathbf{y})}{|\mathbf{x} - \mathbf{y}|^3}. \quad (\text{B.16})$$

This expression makes the connection with flux tube linking explicit [5].

C Helicity densities for moving particles

For point particles with worldlines $z_A^\mu(t) = (t, \mathbf{y}_A(t))$, the stress tensor on a Minkowski background is

$$\begin{aligned} T^{\mu\nu}(t, \mathbf{x}) &= \sum_A m_A \int d\tau_A u_A^\mu u_A^\nu \delta^{(4)}(x - z_A(\tau_A)) \\ &= \sum_A \frac{m_A u_A^\mu u_A^\nu}{u_A^0} \delta^{(3)}(\mathbf{x} - \mathbf{y}_A(t)), \end{aligned} \quad (\text{C.1})$$

where the four-velocity is $u_A^\mu = \gamma_A(1, \mathbf{v}_A)$ and $\gamma_A = (1 - \mathbf{v}_A^2)^{-1/2}$. Thus

$$T^{00} = \sum_A m_A \gamma_A \delta_A, \quad T^{0i} = \sum_A m_A \gamma_A v_A^i \delta_A, \quad T^{ij} = \sum_A m_A \gamma_A v_A^i v_A^j \delta_A, \quad (\text{C.2})$$

with $\delta_A = \delta^{(3)}(\mathbf{x} - \mathbf{y}_A)$. According to the formula (5.5a),

$$\Xi_i = 2G \int d^3 \mathbf{x}' \frac{\delta_{ij} + N_i N_j}{R} T_{0j}(t, \mathbf{x}') = 2G \sum_A \frac{p_{Ai} + N_{Ai} p_{Aj} N_{Aj}}{R_A} \quad (\text{C.3})$$

where

$$p_{Ai} = -m_A \gamma_A v_{Ai}, \quad R_A = |\mathbf{x} - \mathbf{y}_A(t)|, \quad \mathbf{N}_A = \frac{\mathbf{x} - \mathbf{y}_A(t)}{R_A}. \quad (\text{C.4})$$

Here $\mathbf{N}_A$ is the unit vector pointing from particle A to the field point, and R_A is the corresponding distance. This expression should be understood as the leading post-Newtonian term of the spin-1 field, rather than as a complete solution sourced by the matter stress tensor alone. In the conserved effective source of the post-Minkowskian formulation,

$$\tau^{\mu\nu} = \sqrt{-g}T_{\text{pp}}^{\mu\nu} + t_{\text{grav}}^{\mu\nu}, \quad \partial_\mu \tau^{\mu\nu} = 0, \quad (\text{C.5})$$

the leading momentum density is supplied by the point particle matter current, whereas interaction and gravitational field corrections enter at higher post-Newtonian orders. Consequently, the above expression is sufficient for the leading spin-1 helicity densities of a slowly moving binary away from the particle worldlines.

Therefore, the spin-1 gravitomagnetic and gravito-current helicity densities are

$$\mathcal{H}_{NY}^V = -2G^2 \sum_{A \neq B} \frac{[\mathbf{p}_A + \mathbf{N}_A(\mathbf{p}_A \cdot \mathbf{N}_A)] \cdot (\mathbf{p}_B \times \mathbf{N}_B)}{R_A R_B^2}, \quad (\text{C.6a})$$

$$\mathcal{H}_{CS}^V = -8G^2 \sum_{A \neq B} \frac{(\mathbf{p}_A \times \mathbf{N}_A) \cdot [\mathbf{p}_B - 3\mathbf{N}_B(\mathbf{p}_B \cdot \mathbf{N}_B)]}{R_A^2 R_B^3}. \quad (\text{C.6b})$$

The summation is over $A \neq B$. The $A = B$ self-contribution vanishes algebraically at every regular field point. In other words, nonzero terms are interference terms between distinct particle fields. At the worldlines the point-particle fields are singular, so an integrated helicity requires excision, a finite-size model, or a regularization prescription. One can check that the point-particle expression (C.6) for the spin-1 constraint field is consistent with the binary calculation at leading post-Newtonian order.

For $r \gg |\mathbf{y}_A(t)|$, the large- r expansion of Eqs. (C.6a)–(C.6b) reproduces the general asymptotic form (5.21) with the identifications

$$D_{\langle ij \rangle} = \sum_A y_{A\langle i} p_{A j \rangle}, \quad P_i = \sum_A p_{Ai}, \quad S_i = \sum_A \epsilon_{ijk} y_{Aj} p_{Ak}. \quad (\text{C.7})$$

Next we turn to the radiative modes. In the wave zone, where r and t are large while the retarded time $u = t - r$ is held fixed, the transverse traceless metric perturbation takes the form

$$h_{ij}^{\text{TT}} = \frac{4G}{r} \Pi_{ij,kl}(\mathbf{n}) \sum_A \left. \frac{m_A \gamma_A^{\nu} v_{Ak} v_{Al}}{1 - \mathbf{n} \cdot \mathbf{v}_A} \right|_{t_A} + \mathcal{O}(r^{-2}). \quad (\text{C.8})$$

where $t_A = u + \mathbf{n} \cdot \mathbf{y}_A(t_A)$. Above equation is the leading matter contribution to the far-zone TT field. However, its interpretation requires some care. The TT wave equation is not independent of the remaining linearized Einstein equations. By the linearized Bianchi identity, a consistent source must satisfy

$$\partial_\mu T^{\mu\nu} = 0. \quad (\text{C.9})$$

For the point particle matter stress tensor one instead finds

$$\partial_\mu T_{\text{pp}}^{\mu\nu} = \sum_A m_A \int d\tau_A a_A^\nu \delta^{(4)}(x - z_A(\tau_A)) \quad (\text{C.10})$$

where a_A is the acceleration of the particle A . Consequently, Eq. (C.8) can be used directly for freely moving particles in Minkowski spacetime. It can also describe the asymptotic incoming and outgoing inertial states of a scattering process. In such regions the velocities are constant, and the resulting field is an asymptotic boosted Coulomb or memory field rather than radiation produced during the interaction. For a self-gravitating binary, the role of the conserved source is played by

the post-Minkowskian effective stress tensor (C.5) where $t_{\text{grav}}^{\mu\nu}$ contains the gravitational interaction terms. At leading post-Newtonian order, source conservation converts the far-zone stress integral into the second derivative of the mass quadrupole,

$$h_{ij}^{\text{TT}} = \frac{2G}{r} \Pi_{ij,kl}(\mathbf{n}) \ddot{I}_{kl}(u) + \mathcal{O}(r^{-2}, \text{higher PN}). \quad (\text{C.11})$$

Accordingly, the spin-2 helicity densities of a bound binary must be computed from the quadrupole waveform rather than from Eq. (C.8). More precisely, according to Eq.(5.17),

$$h_{ij}^{\text{TT}} = \frac{4G}{r} \Pi_{ij,kl}(\mathbf{n}) \sum_A m_A \left(v_{A\langle k} v_{A\rangle l} + y_{A\langle k} a_{A\rangle l} \right) + \mathcal{O}(r^{-2}, \text{higher PN}). \quad (\text{C.12})$$

In the slow motion limit, the above expression distinguishes from (C.8) by the term associated with the acceleration. With this expression, we find

$$\mathcal{H}_{\text{NY}}^T = \frac{4G^2}{r^2} \sum_{A,B} m_A m_B \left(3v_{A\langle i} a_{A\rangle j} + y_{A\langle i} j_{A\rangle j} \right) \left(v_{B\langle k} v_{B\rangle l} + y_{B\langle k} a_{B\rangle l} \right) Q_{ijkl}(\mathbf{n}) + \mathcal{O}(r^{-3}), \quad (\text{C.13a})$$

$$\mathcal{H}_{\text{CS}}^T = \frac{16G^2}{r^2} \sum_{A,B} m_A m_B \left(3a_{A\langle i} a_{A\rangle j} + 4v_{A\langle i} j_{A\rangle j} + y_{A\langle i} s_{A\rangle j} \right) \left(3v_{B\langle k} a_{B\rangle l} + y_{B\langle k} j_{B\rangle l} \right) Q_{ijkl}(\mathbf{n}) + \mathcal{O}(r^{-3}). \quad (\text{C.13b})$$

In the quadrupole formulas above, which are derived from $I_{ij}(u)$ in Eq. (5.17), all worldline quantities $\mathbf{y}_A, \mathbf{v}_A, \mathbf{a}_A, \mathbf{j}_A, \mathbf{s}_A$, and likewise those with label B , are evaluated at the common retarded time u . Here $j_{Ai} = \dot{a}_{Ai}$ is the jerk, and $s_{Ai} = \dot{j}_{Ai}$ is the snap of particle A , where the dot denotes d/du in these formulas. The particle dependent retarded times t_A defined below Eq. (C.8) belong to the unexpanded matter field expression and are not used in the quadrupole representation.

References

- [1] E. Wigner, “On Unitary Representations of the Inhomogeneous Lorentz Group,” *Annals of Mathematics* **40** (Jan., 1939) 149–204.
- [2] M. G. Calkin, “An invariance property of the free electromagnetic field,” *American Journal of Physics* **33** (1965) 958–960.
- [3] S. Deser and C. Teitelboim, “Duality transformations of abelian and non-abelian gauge fields,” *Physical Review D* **13** (1976) 1592–1597.
- [4] J.-J. Moreau, “Constantes d’un îlot tourbillonnaire en fluide parfait barotrope,” *Comptes Rendus de l’Académie des Sciences de Paris* **252** (1961) 2810–2812.
- [5] H. K. Moffatt, “The degree of knottedness of tangled vortex lines,” *Journal of Fluid Mechanics* **35** (1969), no. 1, 117–129.
- [6] W. M. Elsasser, “Hydromagnetic dynamo theory,” *Reviews of Modern Physics* **28** (Apr., 1956) 135–163.
- [7] L. Woltjer, “A theorem on force-free magnetic fields,” *Proceedings of the National Academy of Sciences* **44** (June, 1958) 489–491.
- [8] E. G. Blackman, “Magnetic Helicity and Large Scale Magnetic Fields: A Primer,” *Space Sci. Rev.* **188** (2015), no. 1-4, 59–91, [1402.0933](#).

- [9] M. A. Berger and G. B. Field, “The topological properties of magnetic helicity,” *Journal of Fluid Mechanics* **147** (1984) 133–148.
- [10] B. Mashhoon, “Gravitoelectromagnetism: A Brief Review,” *arXiv e-prints* (Nov., 2003) gr-qc/0311030, [gr-qc/0311030](#).
- [11] L. F. O. Costa and J. Natario, “Gravito-electromagnetic analogies,” *Gen. Rel. Grav.* **46** (2014) 1792, [1207.0465](#).
- [12] D. Bini, B. Mashhoon, and Y. N. Obukhov, “Gravitomagnetic helicity,” *Phys. Rev. D* **105** (2022), no. 6, 064028, [2112.07550](#).
- [13] M. Henneaux and C. Teitelboim, “Duality in linearized gravity,” *Physical Review D* **71** (2005) 024018, [gr-qc/0408101](#).
- [14] S. M. Barnett, “Maxwellian theory of gravitational waves and their mechanical properties,” *New Journal of Physics* **16** (2014) 023027.
- [15] S. Aghapour, L. Andersson, and R. Bhattacharyya, “Helicity and spin conservation in Maxwell theory and Linearized Gravity,” [1812.03292](#).
- [16] G. Z. Toth, “Energy–momentum tensor and duality symmetry of linearized gravity in the Fierz formalism,” *Class. Quant. Grav.* **39** (2022), no. 7, 075003, [2108.02124](#).
- [17] D. M. Lipkin, “Existence of a new conservation law in electromagnetic theory,” *Journal of Mathematical Physics* **5** (May, 1964) 696–700.
- [18] R. P. Cameron, S. M. Barnett, and A. M. Yao, “Optical helicity, optical spin and related quantities in electromagnetic theory,” *New Journal of Physics* **14** (2012) 053050.
- [19] T. G. Philbin, “Lipkin’s conservation law, noether’s theorem, and the relation to optical helicity,” *Physical Review A* **87** (2013) 043843, [1303.0687](#).
- [20] S. Alexander and J. Martin, “Birefringent gravitational waves and the consistency check of inflation,” *Phys. Rev. D* **71** (2005) 063526, [hep-th/0410230](#).
- [21] K. Kamada, J. Kume, Y. Yamada, and J. Yokoyama, “Gravitational leptogenesis with kination and gravitational reheating,” *JCAP* **01** (2020) 016, [1911.02657](#).
- [22] Z.-Y. Heng, J. Long, R.-Z. Yu, and X.-H. Zhou, “Electromagnetic helicity flux density for radiative systems,” *Class. Quant. Grav.* **43** (2026), no. 1, 015016, [2507.14966](#).
- [23] W.-B. Liu and J. Long, “Symmetry group at future null infinity II: Vector theory,” *JHEP* **07** (2023) 152, [2304.08347](#).
- [24] W.-B. Liu, J. Long, and X.-H. Zhou, “Electromagnetic helicity flux operators in higher dimensions,” *JHEP* **04** (2025) 026, [2407.20077](#).
- [25] J. Long and X.-H. Zhou, “Reduction of topological invariants on null hypersurfaces,” *JHEP* **01** (2026) 116, [2509.06073](#).
- [26] H. Bondi, M. G. J. van der Burg, and A. W. K. Metzner, “Gravitational waves in general relativity. VII. waves from axisymmetric isolated systems,” *Proceedings of the Royal Society of London A* **269** (1962) 21–52.
- [27] R. K. Sachs, “Gravitational waves in general relativity. VIII. waves in asymptotically flat space-times,” *Proceedings of the Royal Society of London A* **270** (1962) 103–126.

- [28] T. Mädler and J. Winicour, “Bondi–sachs formalism,” *Scholarpedia* **11** (2016) 33528, [1609.01731](#).
- [29] W.-B. Liu and J. Long, “Symmetry group at future null infinity III : gravitational theory,” [2307.01068](#).
- [30] H. T. Nieh and M. L. Yan, “An Identity in Riemann-cartan Geometry,” *J. Math. Phys.* **23** (1982) 373.
- [31] T. Kimura, “Divergence of axial-vector current in the gravitational field,” *Progress of Theoretical Physics* **42** (1969) 1191–1205.
- [32] R. Delbourgo and A. Salam, “The gravitational correction to PCAC,” *Physics Letters B* **40** (1972) 381–382.
- [33] O. Chandia and J. Zanelli, “Topological invariants, instantons and chiral anomaly on spaces with torsion,” *Physical Review D* **55** (1997) 7580, [hep-th/9702025](#).
- [34] Y. N. Obukhov, E. W. Mielke, J. Budczies, and F. W. Hehl, “On the chiral anomaly in non-riemannian space-times,” *Foundations of Physics* **27** (1997) 1221–1236, [gr-qc/9702011](#).
- [35] E. E. Flanagan and S. A. Hughes, “The Basics of gravitational wave theory,” *New J. Phys.* **7** (2005) 204, [gr-qc/0501041](#).
- [36] J. M. Bardeen, “Gauge Invariant Cosmological Perturbations,” *Phys. Rev. D* **22** (1980) 1882–1905.
- [37] H. Kodama and M. Sasaki, “Cosmological Perturbation Theory,” *Prog. Theor. Phys. Suppl.* **78** (1984) 1–166.
- [38] M. Krssak, R. J. van den Hoogen, J. G. Pereira, C. G. Böhrer, and A. A. Coley, “Teleparallel theories of gravity: illuminating a fully invariant approach,” *Classical and Quantum Gravity* **36** (2019) 183001, [1810.12932](#).
- [39] A. Brandenburg and K. Subramanian, “Astrophysical magnetic fields and nonlinear dynamo theory,” *Phys. Rept.* **417** (2005) 1–209, [astro-ph/0405052](#).
- [40] M. Satoh, S. Kanno, and J. Soda, “Circular Polarization of Primordial Gravitational Waves in String-inspired Inflationary Cosmology,” *Phys. Rev. D* **77** (2008) 023526, [0706.3585](#).
- [41] A. Einstein, “Riemann-geometrie mit aufrechterhaltung des begriffes des fernparallelismus,” *Sitzungsberichte der Preussischen Akademie der Wissenschaften, Physikalisch-Mathematische Klasse* (1928) 217–221.
- [42] Y. M. Cho, “Einstein lagrangian as the translational yang–mills lagrangian,” *Physical Review D* **14** (1976) 2521.
- [43] K. Hayashi and T. Nakano, “Extended translation invariance and associated gauge fields,” *Progress of Theoretical Physics* **38** (1967) 491–507.
- [44] K. Hayashi and T. Shirafuji, “New general relativity,” *Physical Review D* **19** (1979) 3524–3553. Addendum: *Phys. Rev. D* **24**, 3312–3314 (1982).
- [45] J. W. Maluf, “The teleparallel equivalent of general relativity,” *Annalen der Physik* **525** (2013) 339–357, [1303.3897](#).

- [46] S. Bahamonde *et al.*, “Teleparallel gravity: from theory to cosmology,” *Reports on Progress in Physics* **86** (2023) 026901, [2106.13793](#).
- [47] M. Krššák and J. G. Pereira, “Spin connection and renormalization of teleparallel action,” *European Physical Journal C* **75** (2015), no. 11, 519, [1504.07683](#).
- [48] M. Maggiore, *Gravitational Waves. Vol. 1: Theory and Experiments*. Oxford University Press, 2007.
- [49] K. S. Thorne, “Multipole Expansions of Gravitational Radiation,” *Rev. Mod. Phys.* **52** (1980) 299–339.
- [50] T. Damour and B. R. Iyer, “Multipole analysis for electromagnetism and linearized gravity with irreducible cartesian tensors,” *Phys. Rev. D* **43** (1991) 3259–3272.
- [51] L. Blanchet, “Gravitational Radiation from Post-Newtonian Sources and Inspiralling Compact Binaries,” *Living Reviews in Relativity* **17** (Dec., 2014) 2, [1310.1528](#).
- [52] R. H. Price, “Nonspherical perturbations of relativistic gravitational collapse. i. scalar and gravitational perturbations,” *Phys. Rev. D* **5** (1972) 2419–2438.
- [53] J. Long and R.-Z. Yu, “Gravitational helicity flux density from two-body systems,” *Class. Quant. Grav.* **42** (2025), no. 4, 045005, [2403.18627](#).
- [54] J. Calderón Bustillo, A. del Rio, N. Sanchis-Gual, K. Chandra, and S. H. W. Leong, “Testing mirror symmetry in the universe with ligo-virgo black-hole mergers,” *Phys. Rev. Lett.* **134** (Jan, 2025) 031402, [2402.09861](#).
- [55] P. C. Peters and J. Mathews, “Gravitational radiation from point masses in a keplerian orbit,” *Phys. Rev.* **131** (1963) 435–439.
- [56] P. C. Peters, “Gravitational radiation and the motion of two point masses,” *Phys. Rev.* **136** (1964) B1224–B1232.
- [57] D. Brouwer and G. M. Clemence, *Methods of Celestial Mechanics*. Academic Press, New York, 1961.
- [58] M. Turner, “Gravitational radiation from point-masses in unbound orbits: Newtonian results,” *Astrophys. J.* **216** (1977) 610–619.
- [59] A. Burinskii and G. Magli, “Kerr-Schild approach to the boosted Kerr solution,” *Phys. Rev. D* **61** (2000) 044017, [gr-qc/9904012](#).
- [60] M. F. Huq, M. W. Choptuik, and R. A. Matzner, “Locating boosted Kerr and Schwarzschild apparent horizons,” *Phys. Rev. D* **66** (2002) 084024, [gr-qc/0002076](#).
- [61] G. C. Debney, R. P. Kerr, and A. Schild, “Solutions of the einstein and einstein-maxwell equations,” *J. Math. Phys.* **10** (1969) 1842–1854.
- [62] H. W. Brinkmann, “Einstein spaces which are mapped conformally on each other,” *Mathematische Annalen* **94** (1925) 119–145.
- [63] W. B. Bonnor, “Spinning null fluid in general relativity,” *International Journal of Theoretical Physics* **3** (1970) 257–266.
- [64] V. P. Frolov and D. V. Fursaev, “Gravitational field of a spinning radiation beam-pulse in higher dimensions,” *Physical Review D* **71** (2005), no. 10, 104034, [hep-th/0504027](#).

- [65] V. P. Frolov, W. Israel, and A. Zelnikov, “Gravitational field of relativistic gyratons,” *Physical Review D* **72** (2005), no. 8, 084031, [hep-th/0506001](#).
- [66] J. Podolský, R. Steinbauer, and R. Švarc, “Gyratonic pp-waves and their impulsive limit,” *Physical Review D* **90** (2014), no. 4, 044050, [1406.3227](#).
- [67] V. Pravda, A. Pravdová, A. Coley, and R. Milson, “All spacetimes with vanishing curvature invariants,” *Classical and Quantum Gravity* **19** (2002), no. 23, 6213–6236, [gr-qc/0209024](#).
- [68] A. Coley, R. Milson, V. Pravda, and A. Pravdová, “Vanishing scalar invariant spacetimes in higher dimensions,” *Classical and Quantum Gravity* **21** (2004), no. 23, 5519–5542, [gr-qc/0410070](#).
- [69] J. Swearngin, A. Thompson, A. Wickes, J. W. Dalhuisen, and D. Bouwmeester, “Gravitational Hopfions,” [1302.1431](#).
- [70] A. Thompson, J. Swearngin, and D. Bouwmeester, “Linked and knotted gravitational radiation,” *Journal of Physics A: Mathematical and Theoretical* **47** (aug, 2014) 355205.
- [71] A. Thompson, A. Wickes, J. Swearngin, and D. Bouwmeester, “Classification of Electromagnetic and Gravitational Hopfions by Algebraic Type,” *J. Phys. A* **48** (2015), no. 20, 205202, [1411.2073](#).
- [72] T. Smolka and J. Jezierski, “Simple description of generalized electromagnetic and gravitational hopfions,” *Class. Quant. Grav.* **35** (2018), no. 24, 245010, [1802.01467](#).
- [73] J. F. Plebanski, “Some solutions of complex Einstein equations,” *J. Math. Phys.* **16** (1975) 2395–2402.
- [74] G. A. J. Sparling and K. P. Tod, “An Example of an H Space,” *J. Math. Phys.* **22** (1981) 331–332.
- [75] K. P. Tod, “Self-dual Kerr–Schild metrics and null Maxwell fields,” *J. Math. Phys.* **23** (1982), no. 6, 1147–1148.
- [76] S. Sabharwal and J. W. Dalhuisen, “Anti-Self-Dual Spacetimes, Gravitational Instantons and Knotted Zeros of the Weyl Tensor,” *JHEP* **07** (2019) 004, [1904.06030](#).
- [77] M. Arrayás, D. Bouwmeester, and J. L. Trueba, “Knots in electromagnetism,” *Phys. Rept.* **667** (2017) 1–61.
- [78] J. Gladikowski and M. Hellmund, “Static solitons with nonzero Hopf number,” *Phys. Rev. D* **56** (1997) 5194–5199, [hep-th/9609035](#).
- [79] H. K. Urbantke, “The Hopf fibration—seven times in physics,” *Journal of Geometry and Physics* **46** (May, 2003) 125–150.
- [80] A. M. Stewart, “Vector potential of the coulomb gauge,” *European Journal of Physics* **24** (2003), no. 5, 519–524.
- [81] K. Y. Bliokh, A. Y. Bekshaev, and F. Nori, “Dual electromagnetism: Helicity, spin, momentum and angular momentum,”